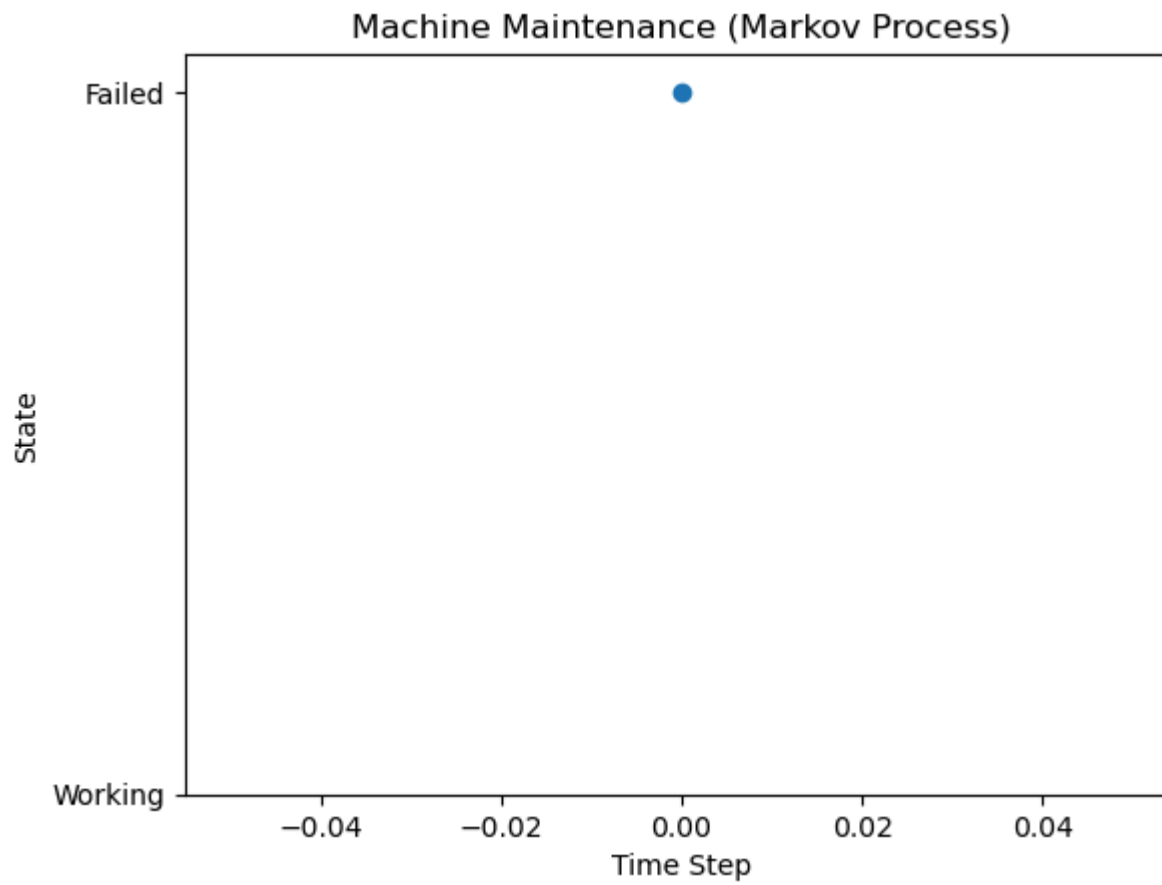


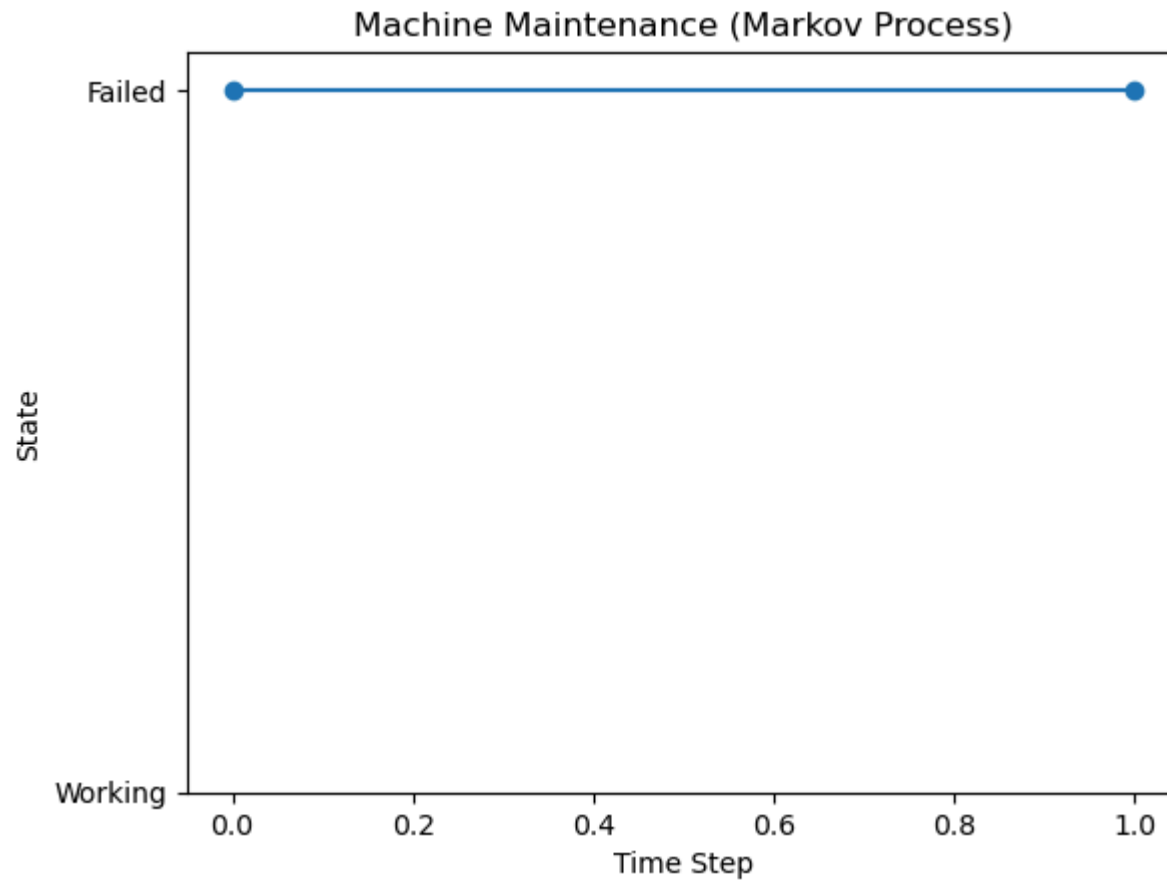
```
In [1]: import numpy as np
import matplotlib.pyplot as plt
transition_matrix = np.array([
    [0.8, 0.2],
    [0.6, 0.4]
])
states = ["Working", "Failed"]
current_state = 0
history = []
plt.ion()
for i in range(50):

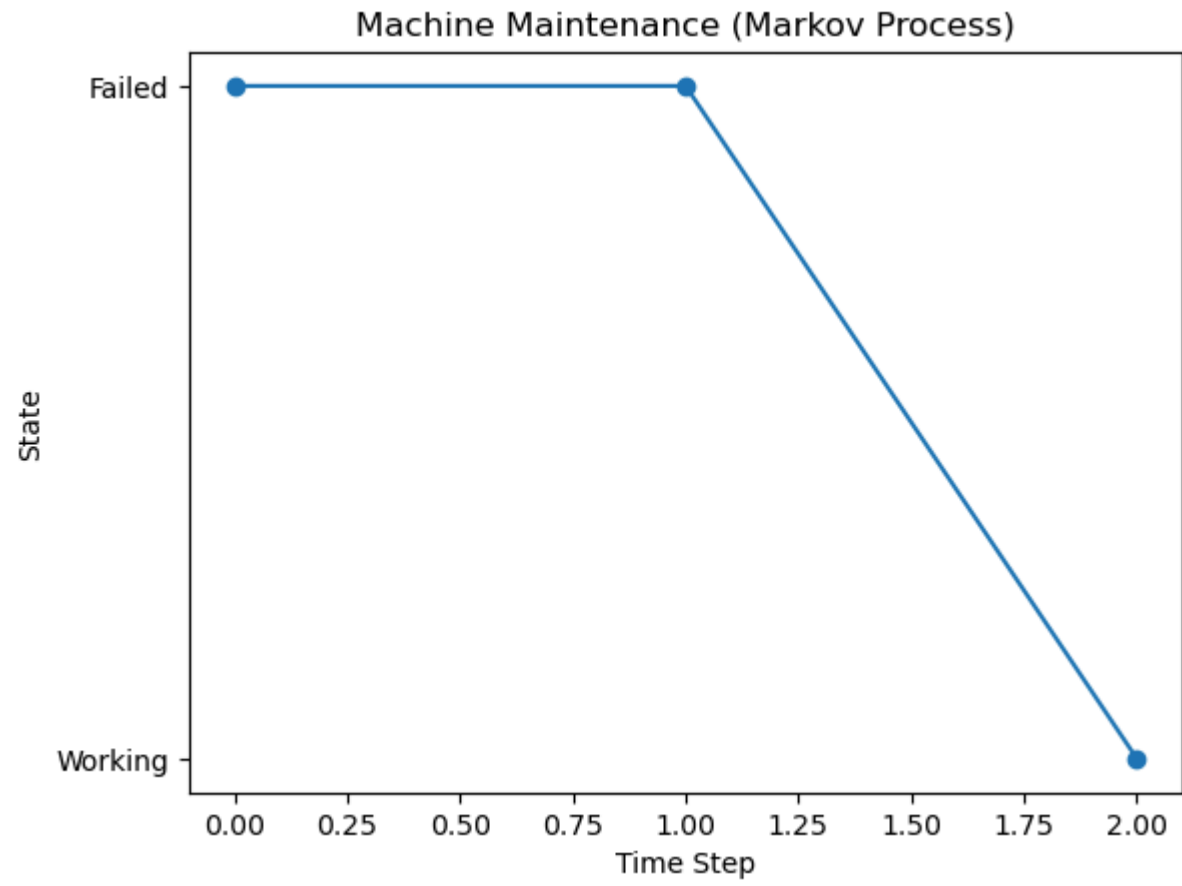
    current_state = np.random.choice([0, 1],
                                     p=transition_matrix[current_state])

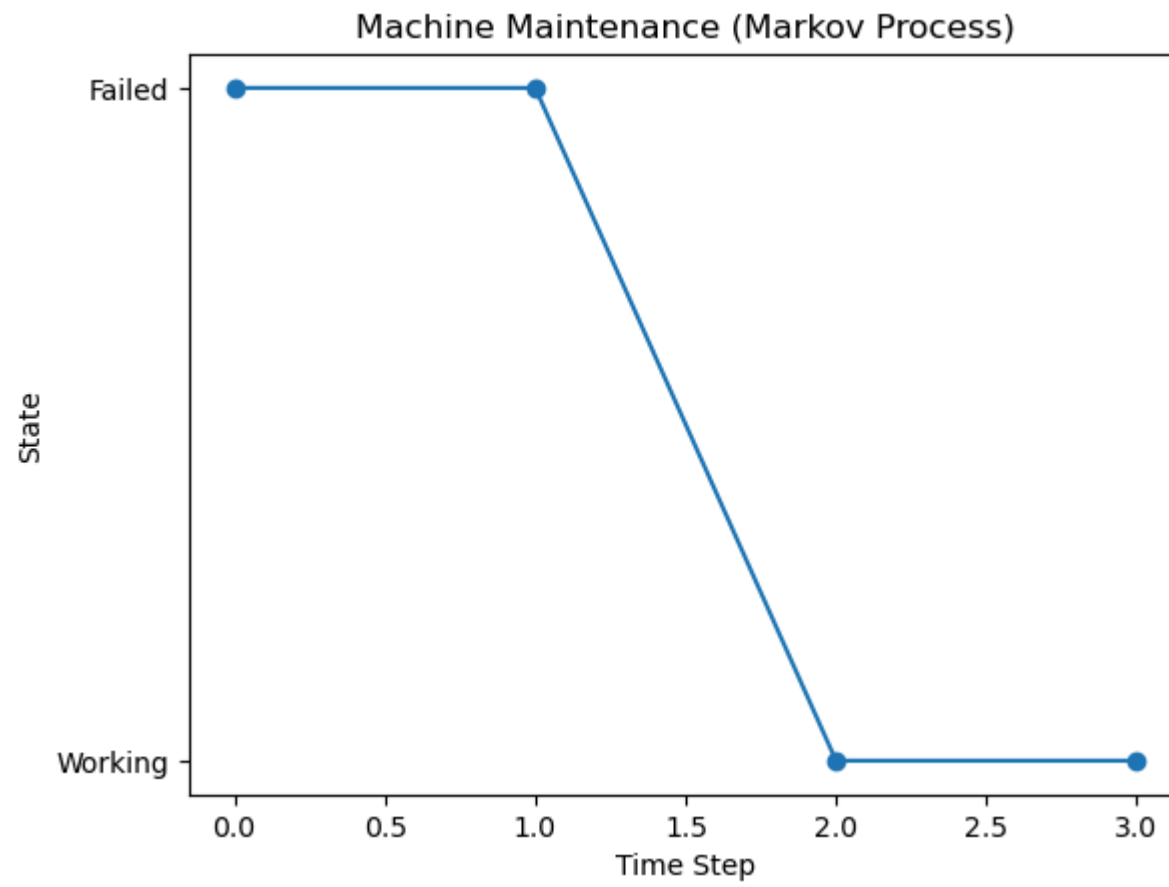
    history.append(current_state)

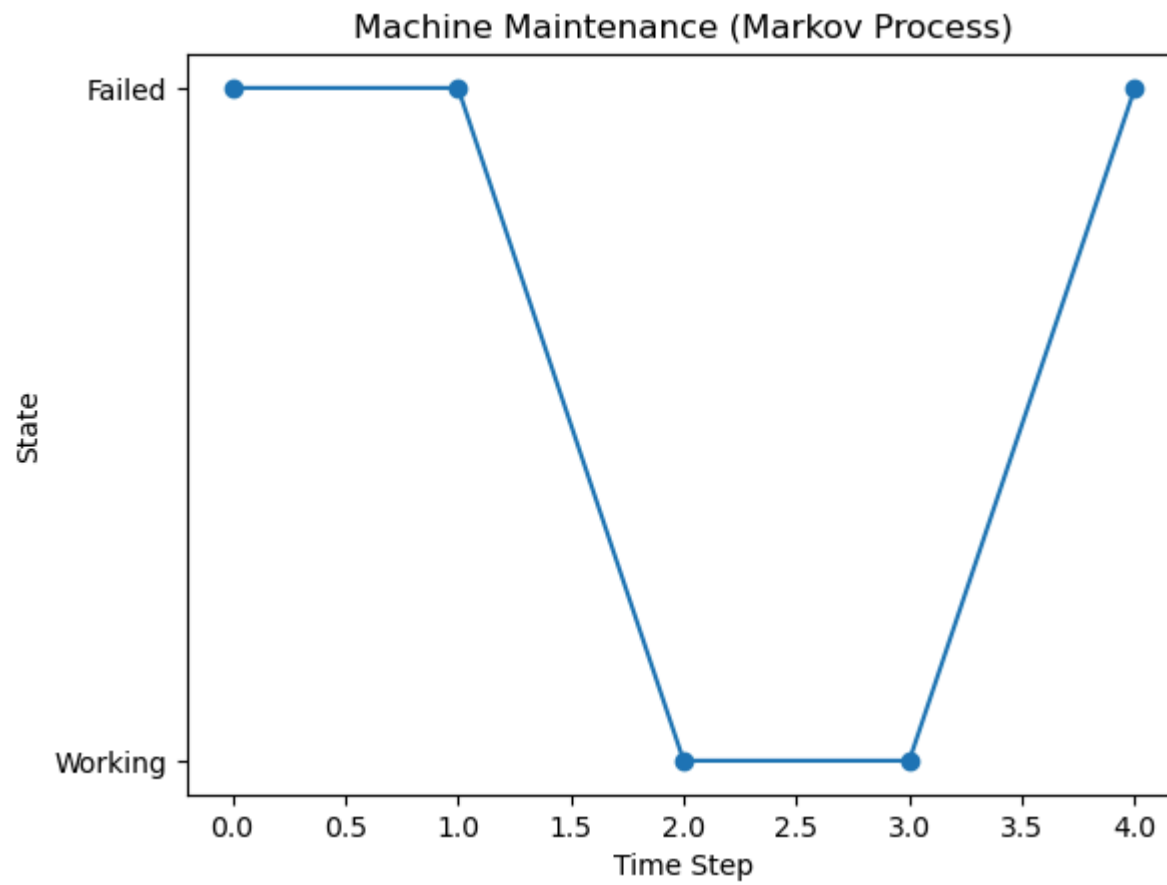
    plt.clf()
    plt.plot(history, marker='o')
    plt.yticks([0, 1], states)
    plt.title("Machine Maintenance (Markov Process)")
    plt.xlabel("Time Step")
    plt.ylabel("State")
    plt.pause(0.2)
plt.ioff()
plt.show()
```

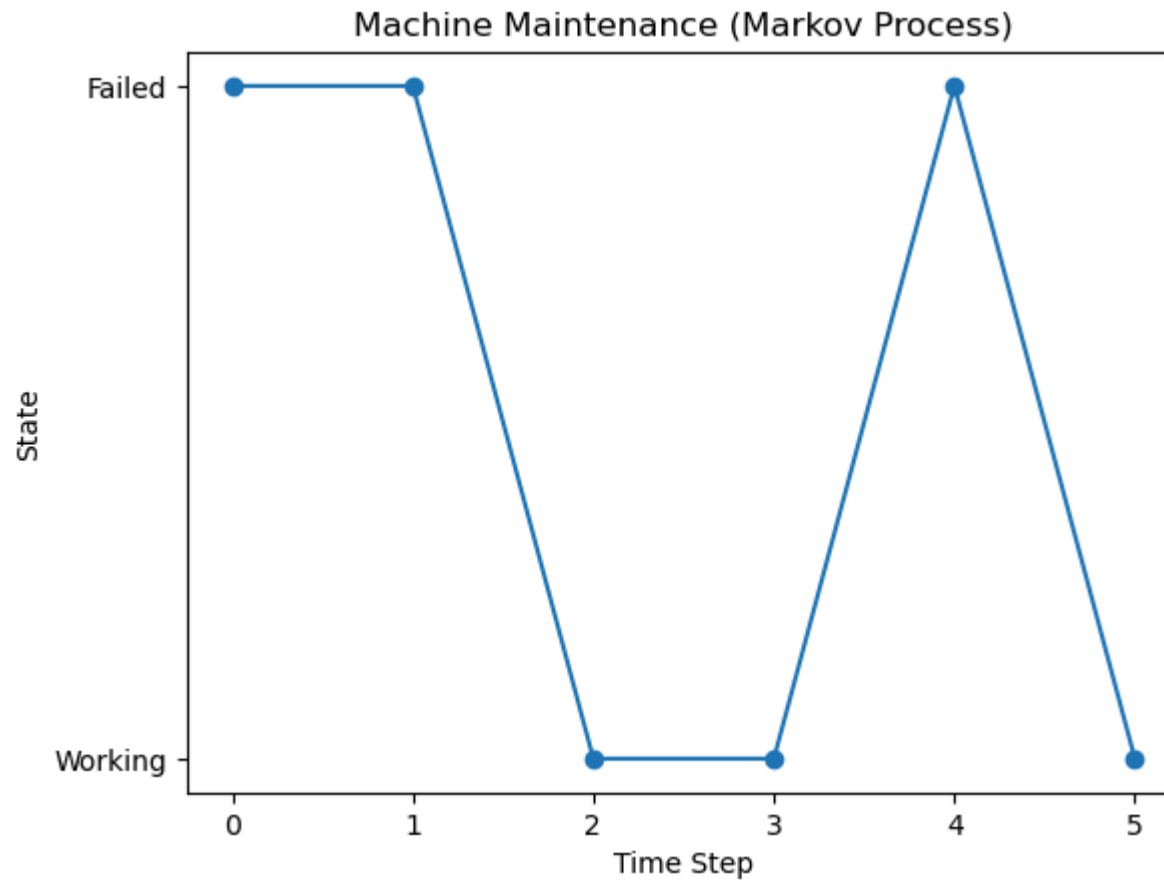


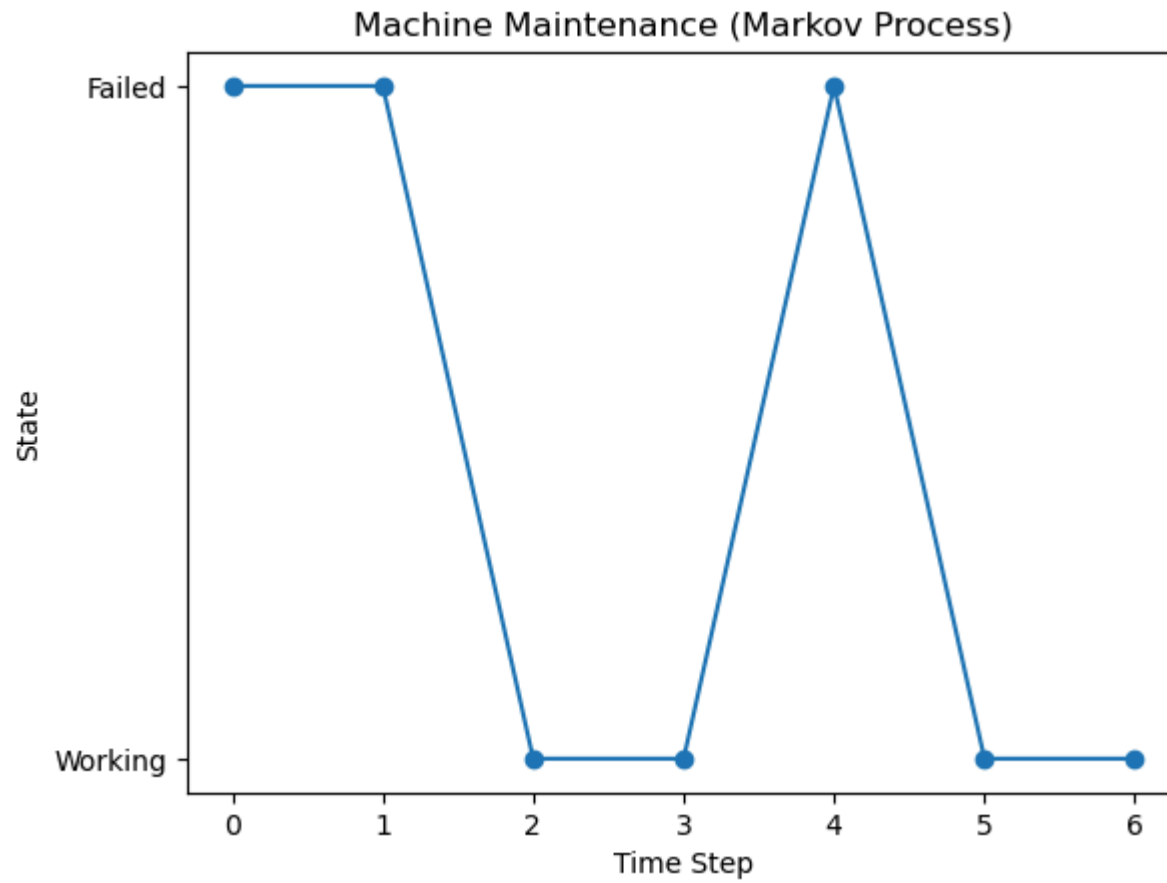


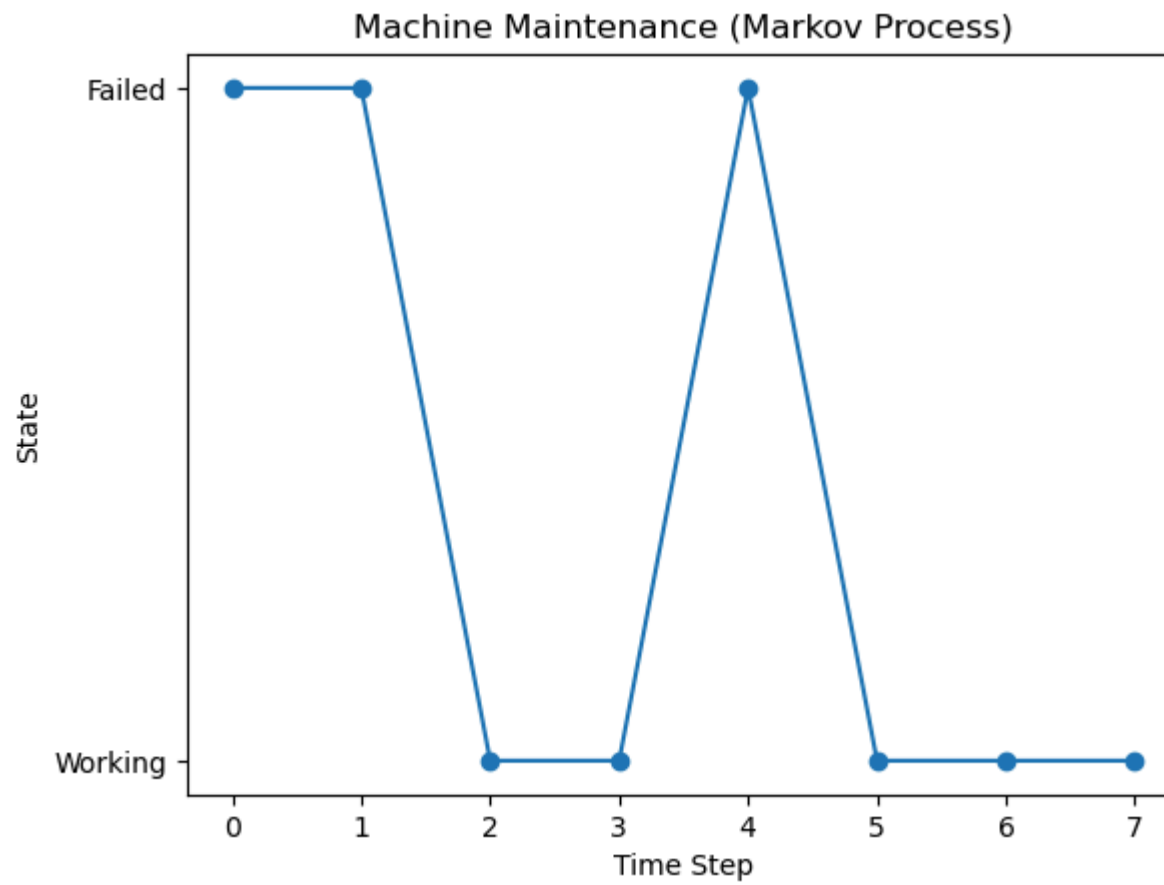


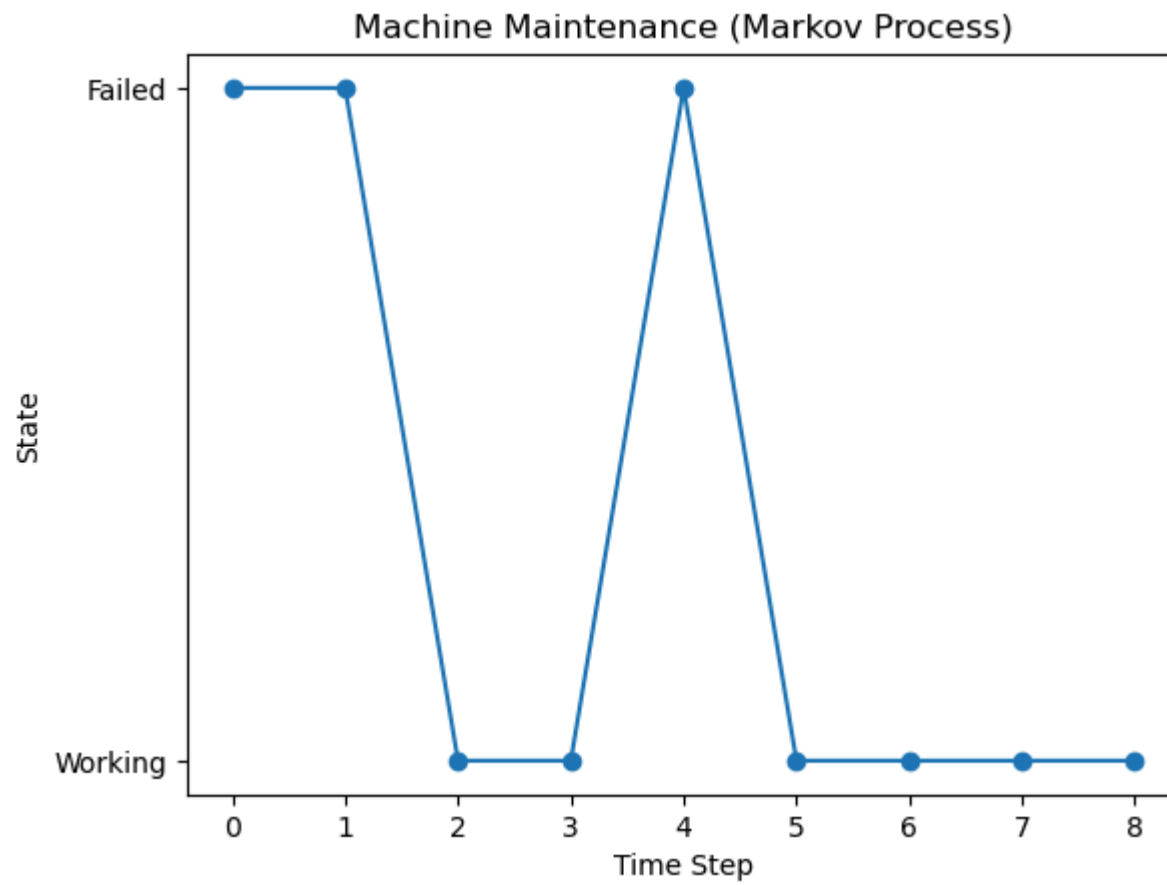


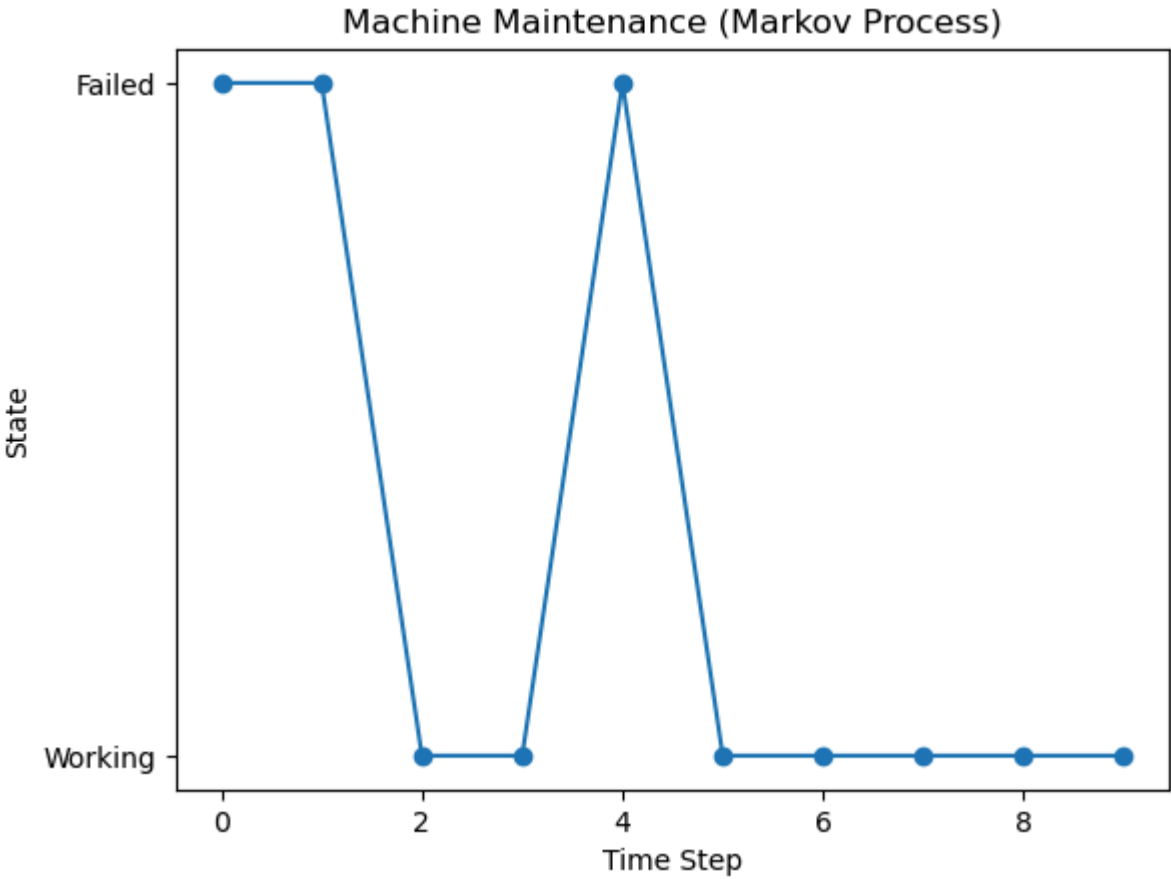


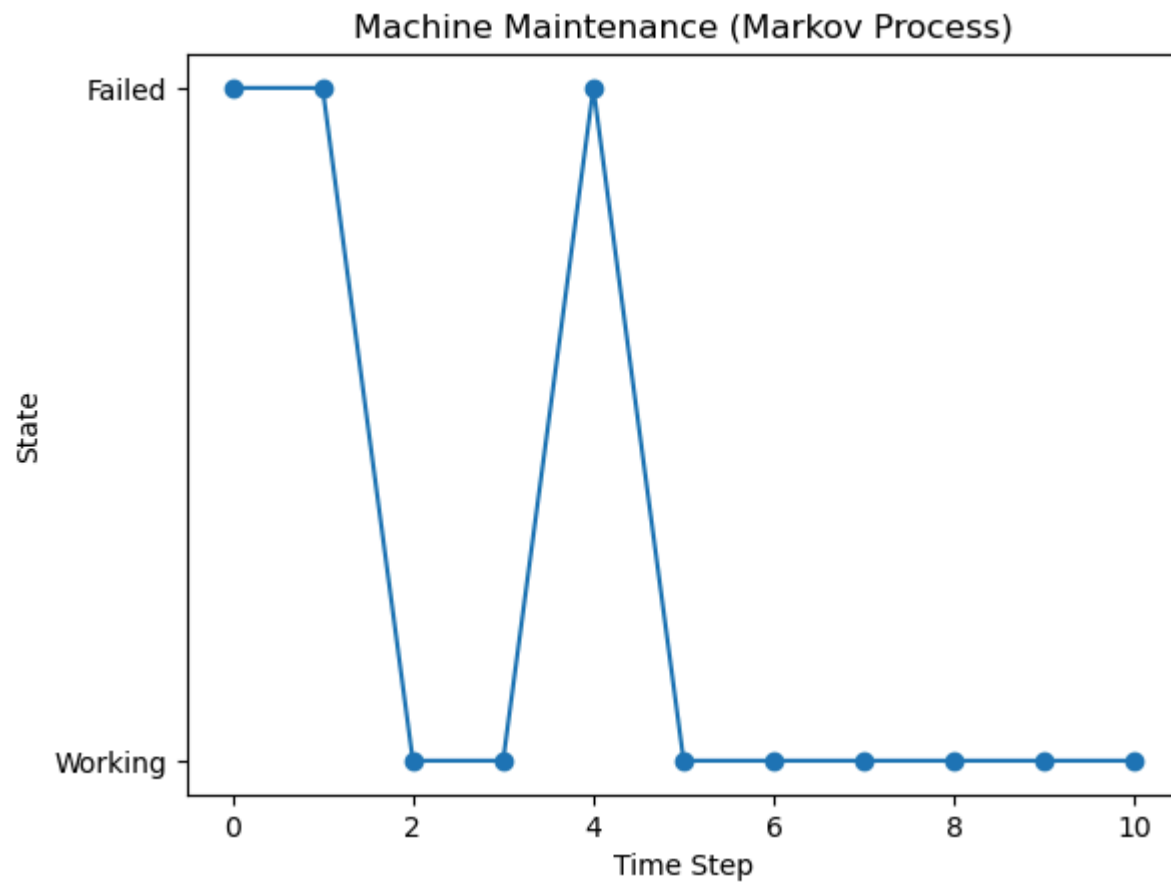


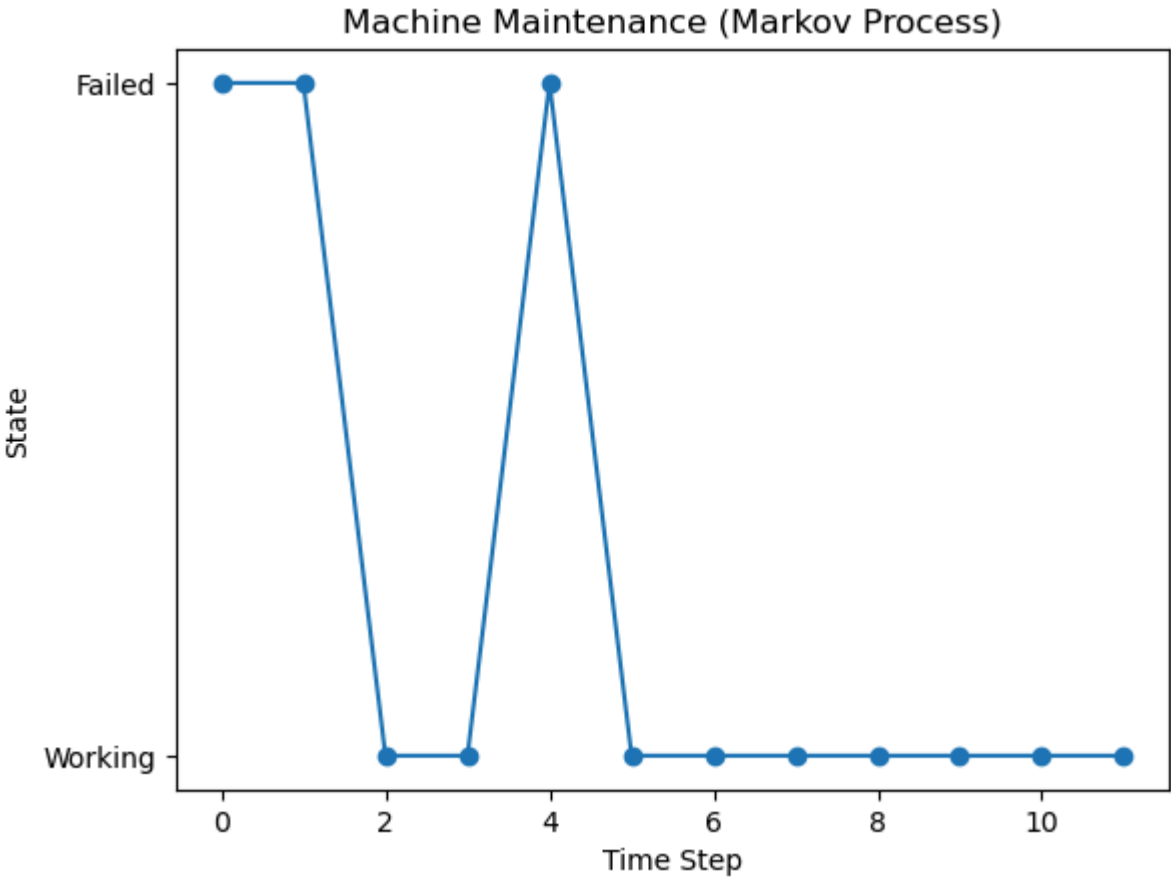


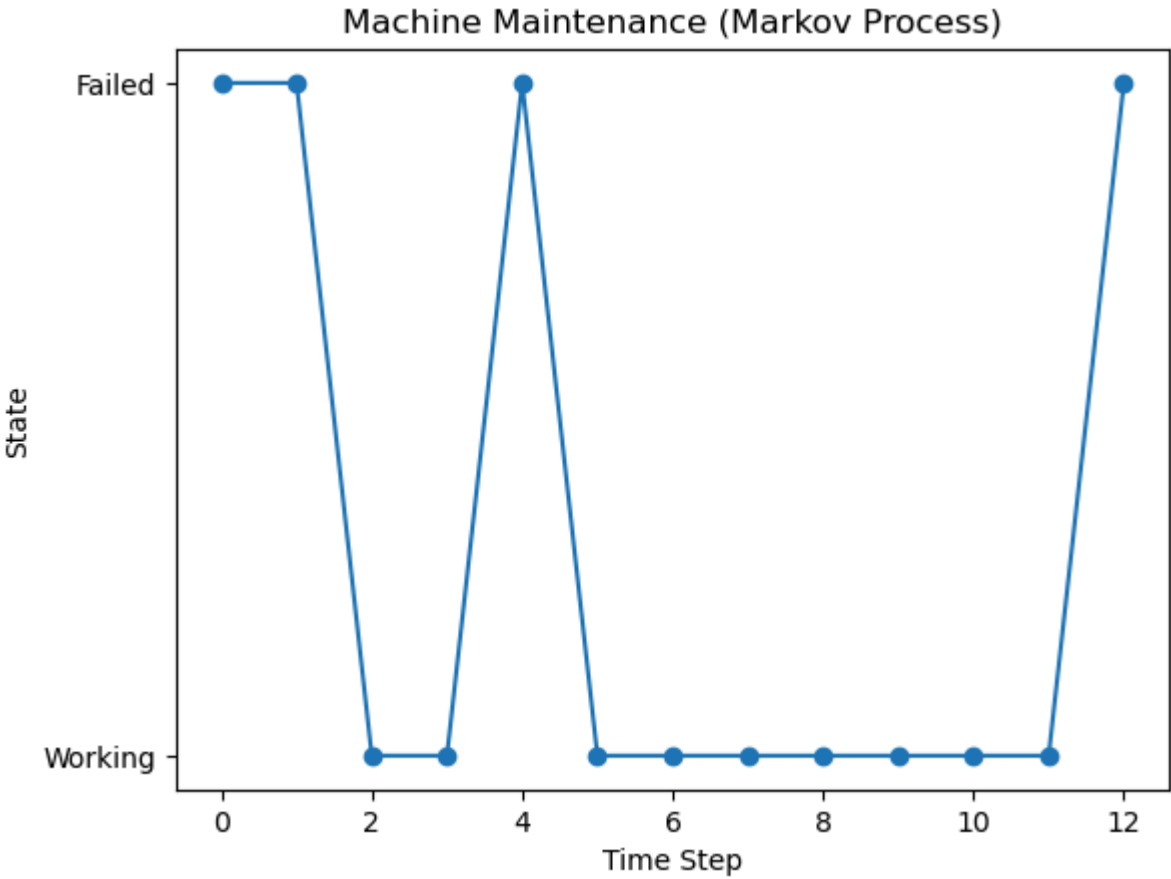


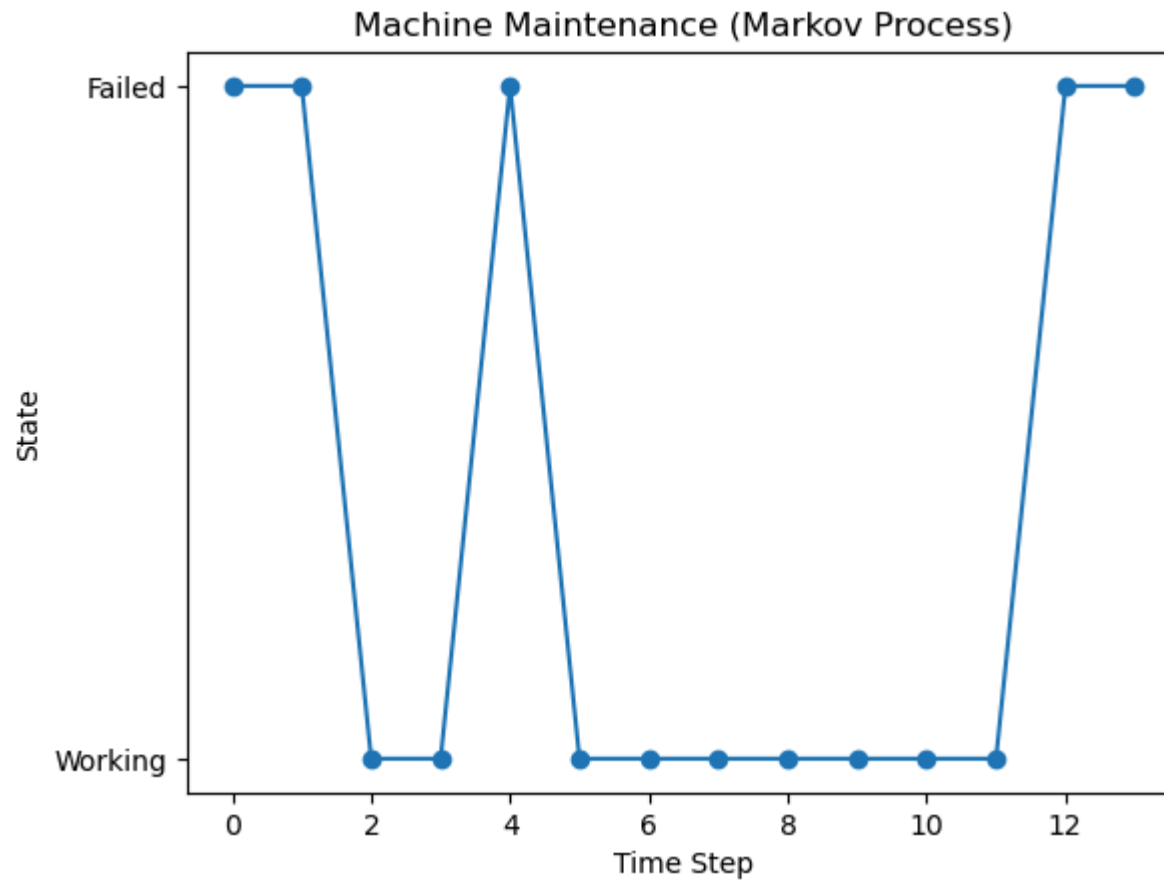


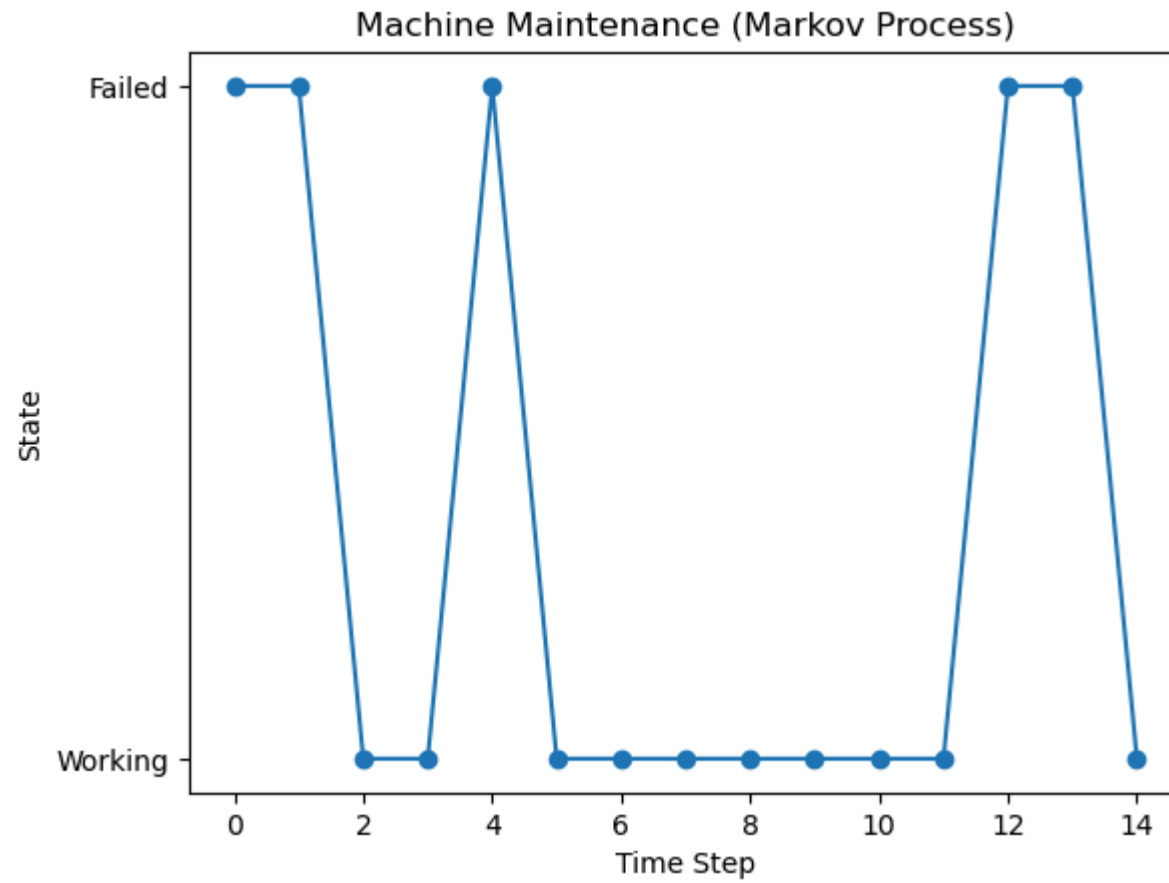


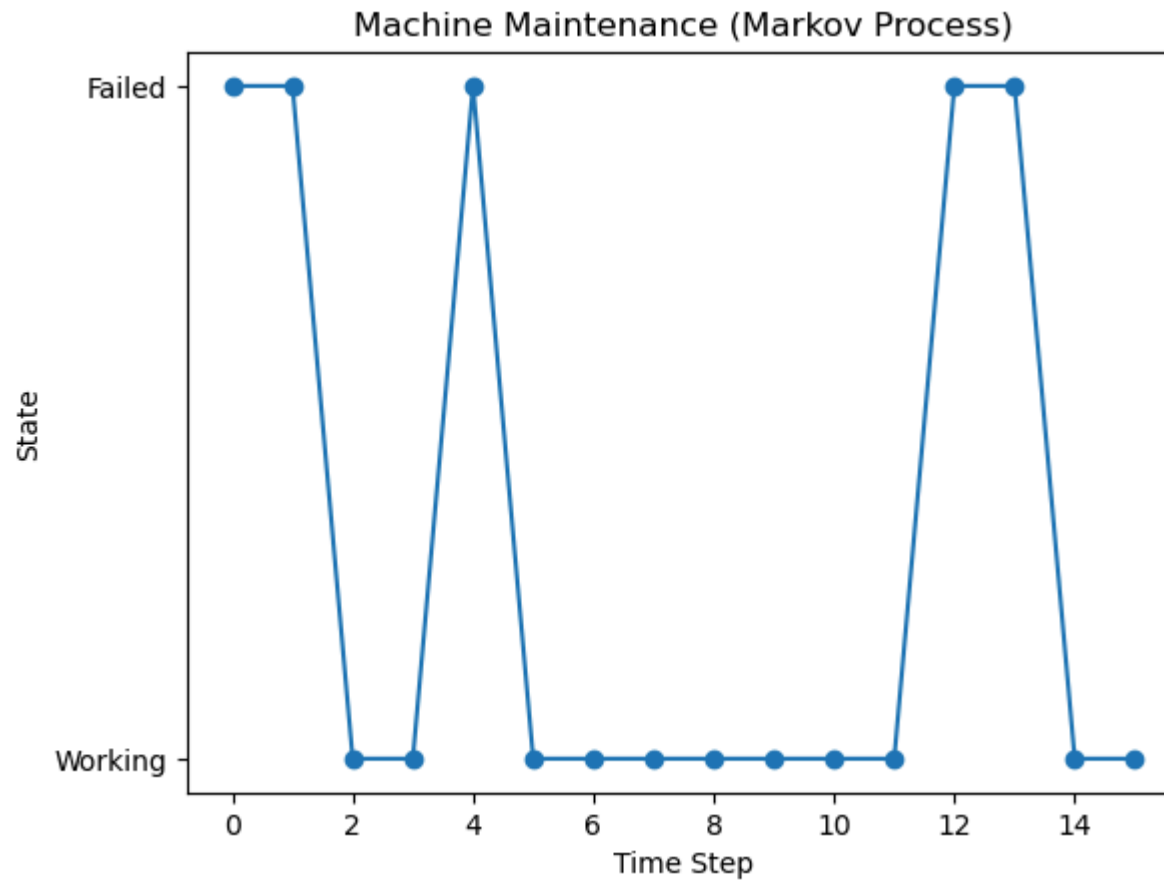


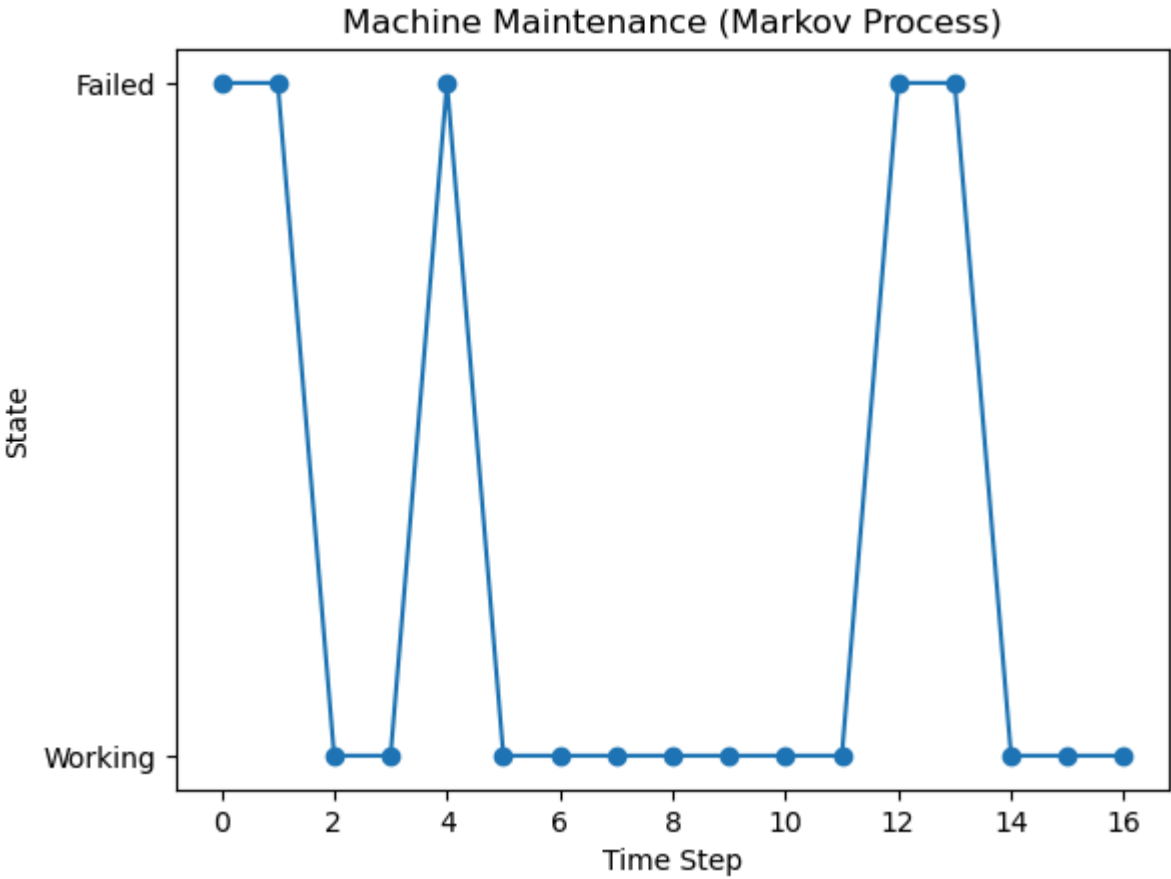


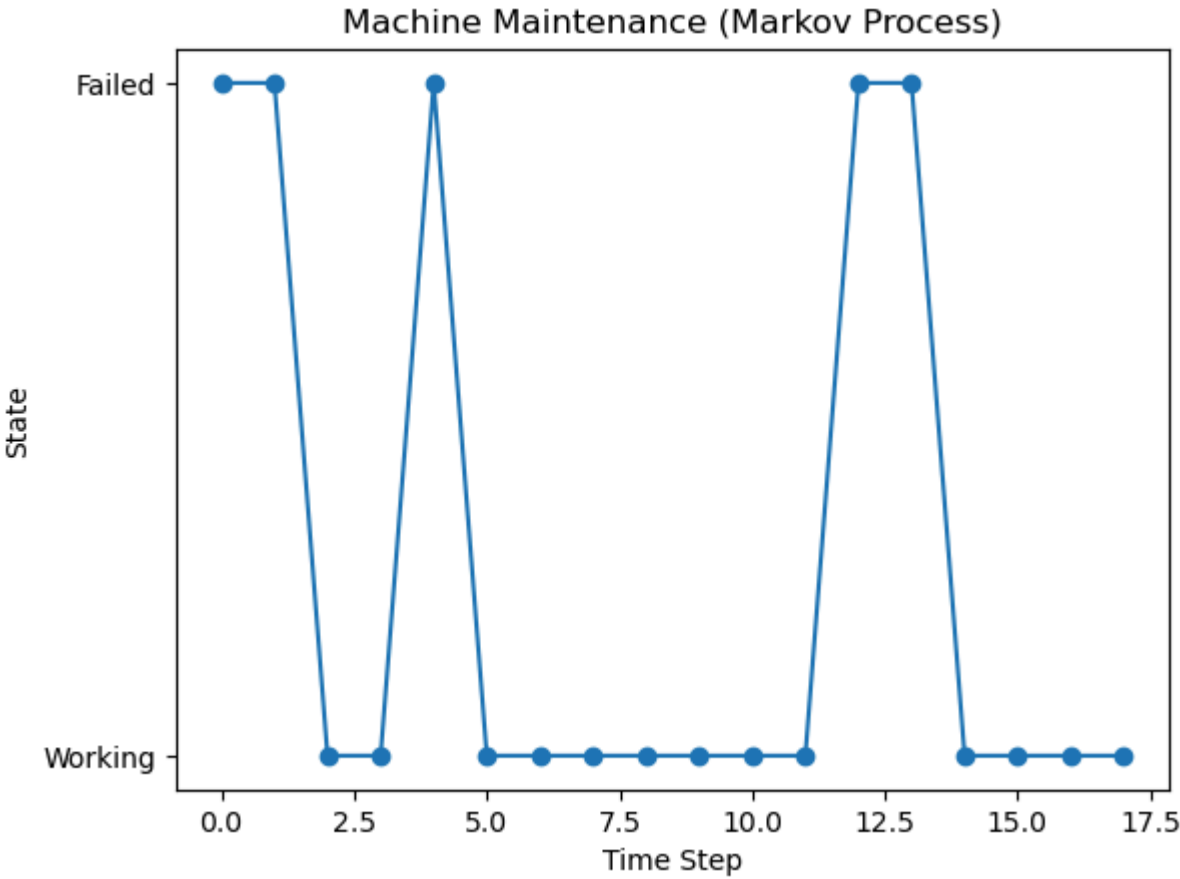


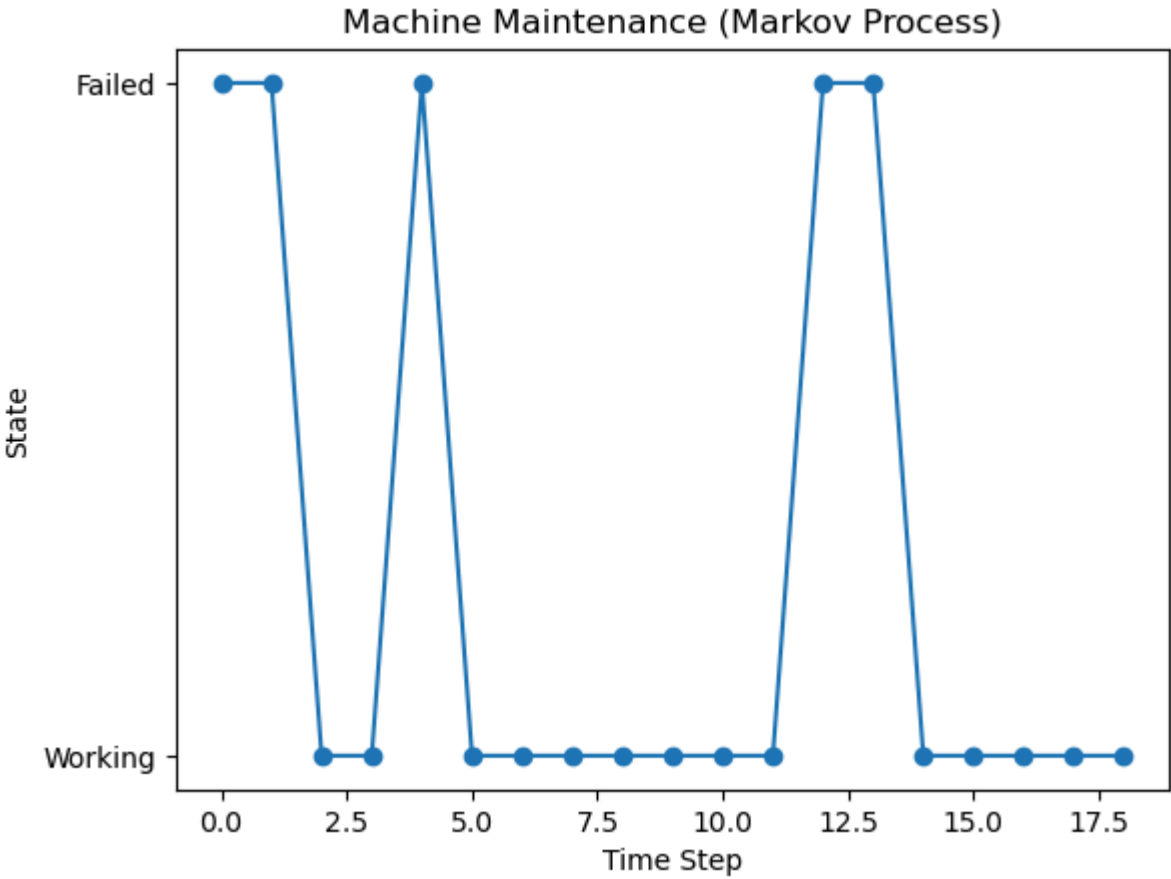


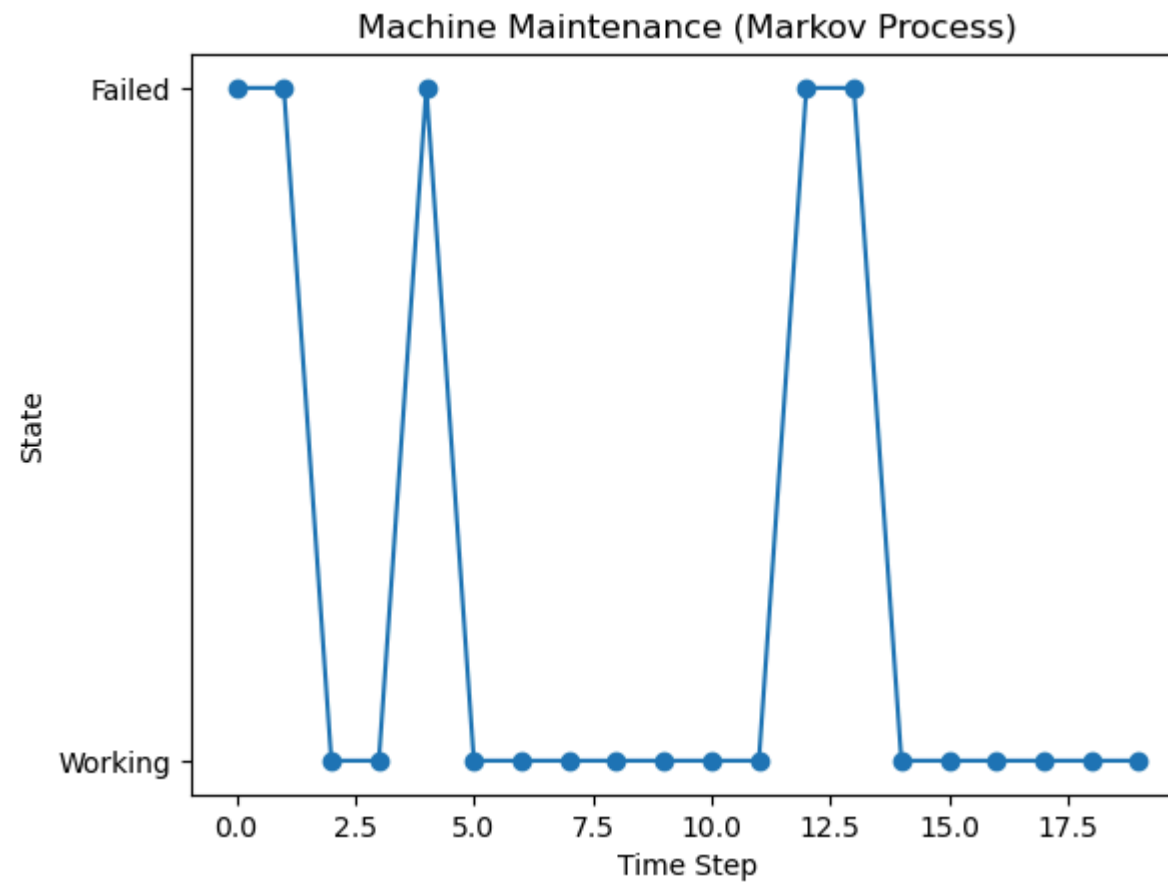


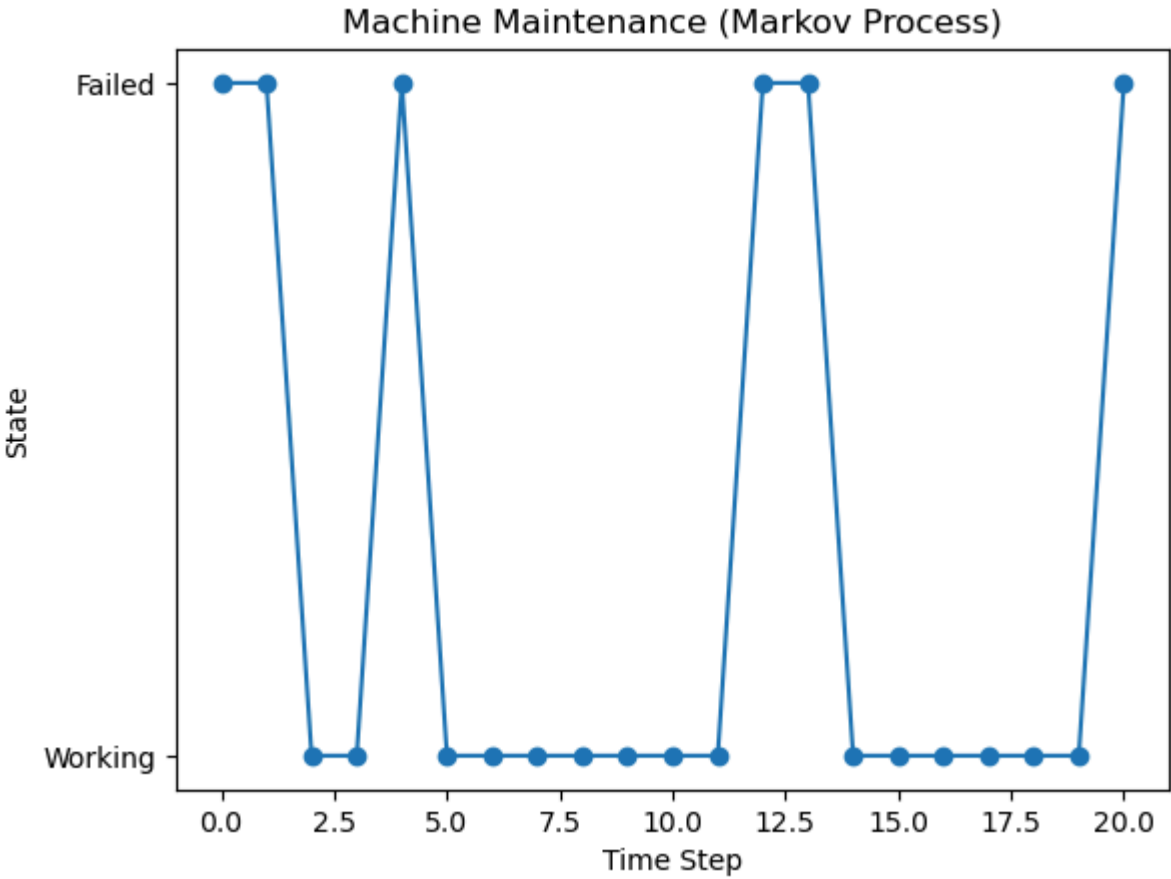


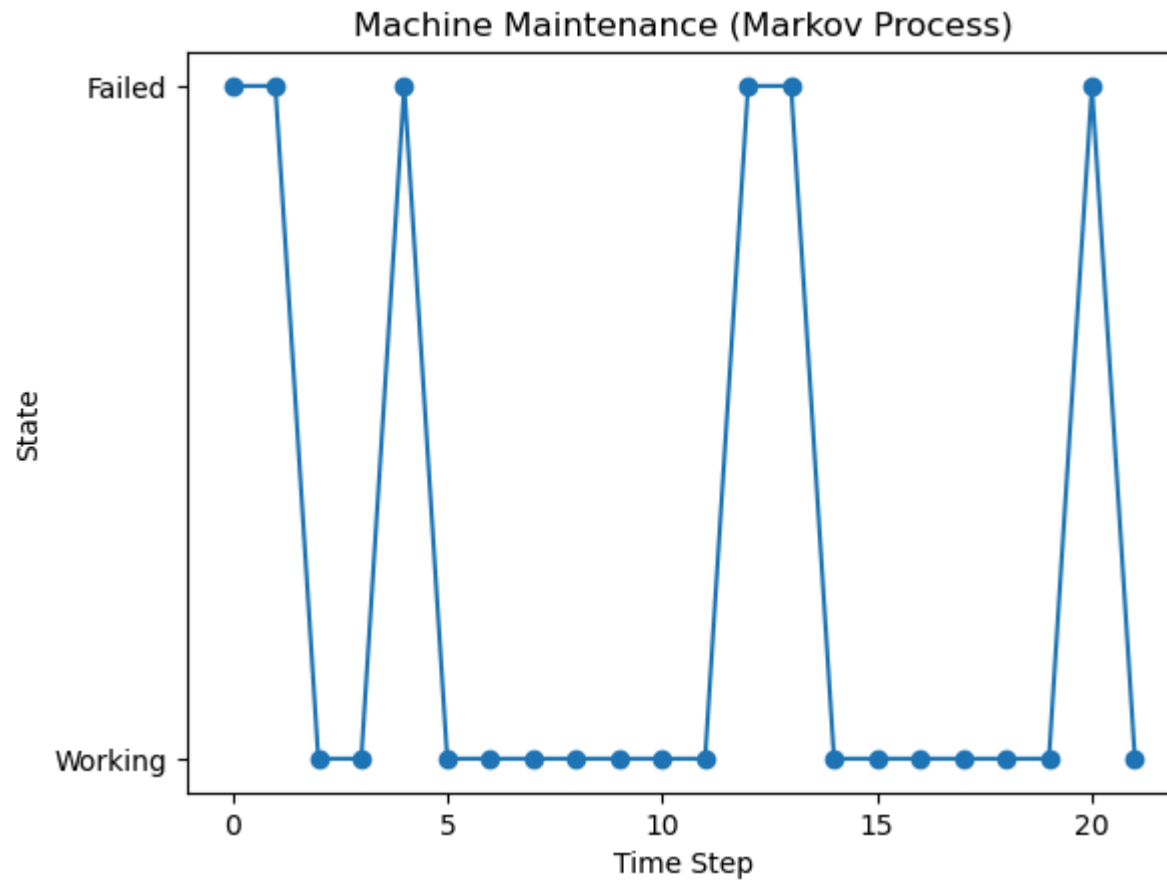


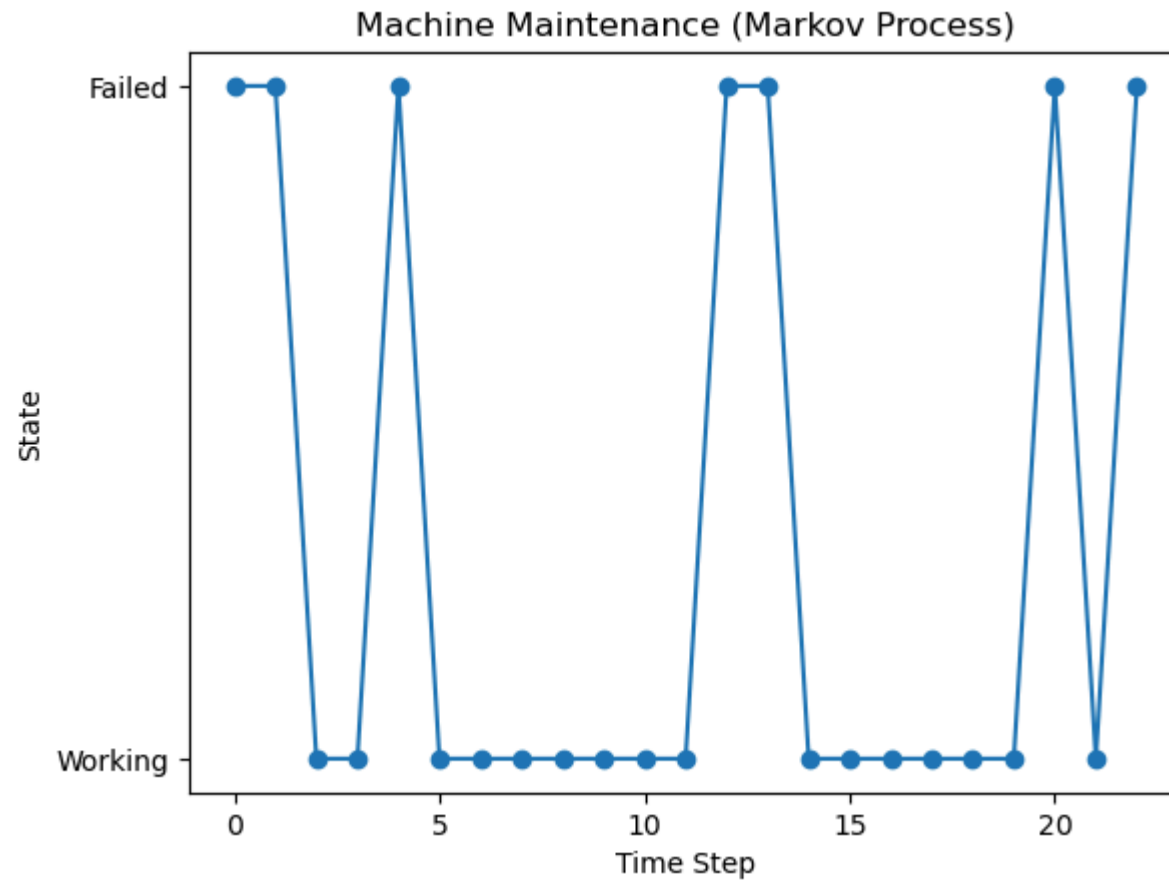


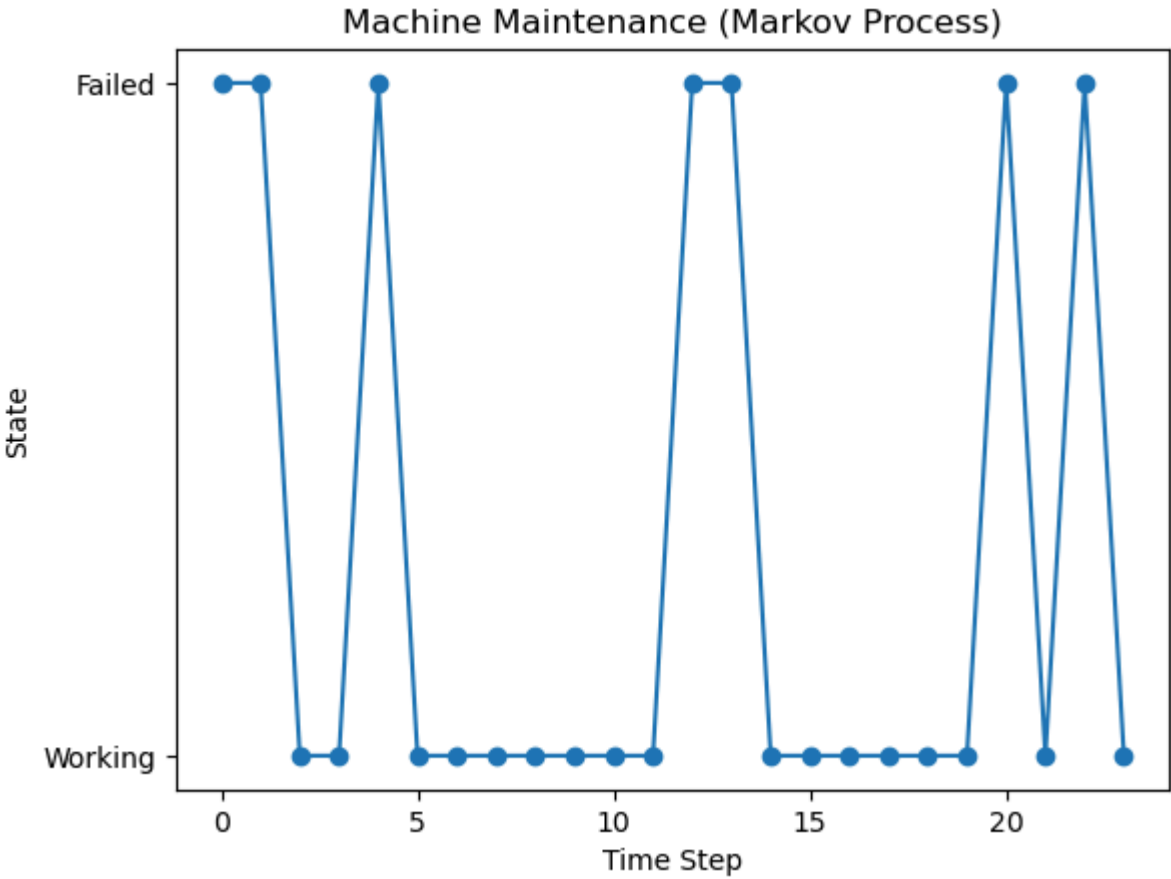


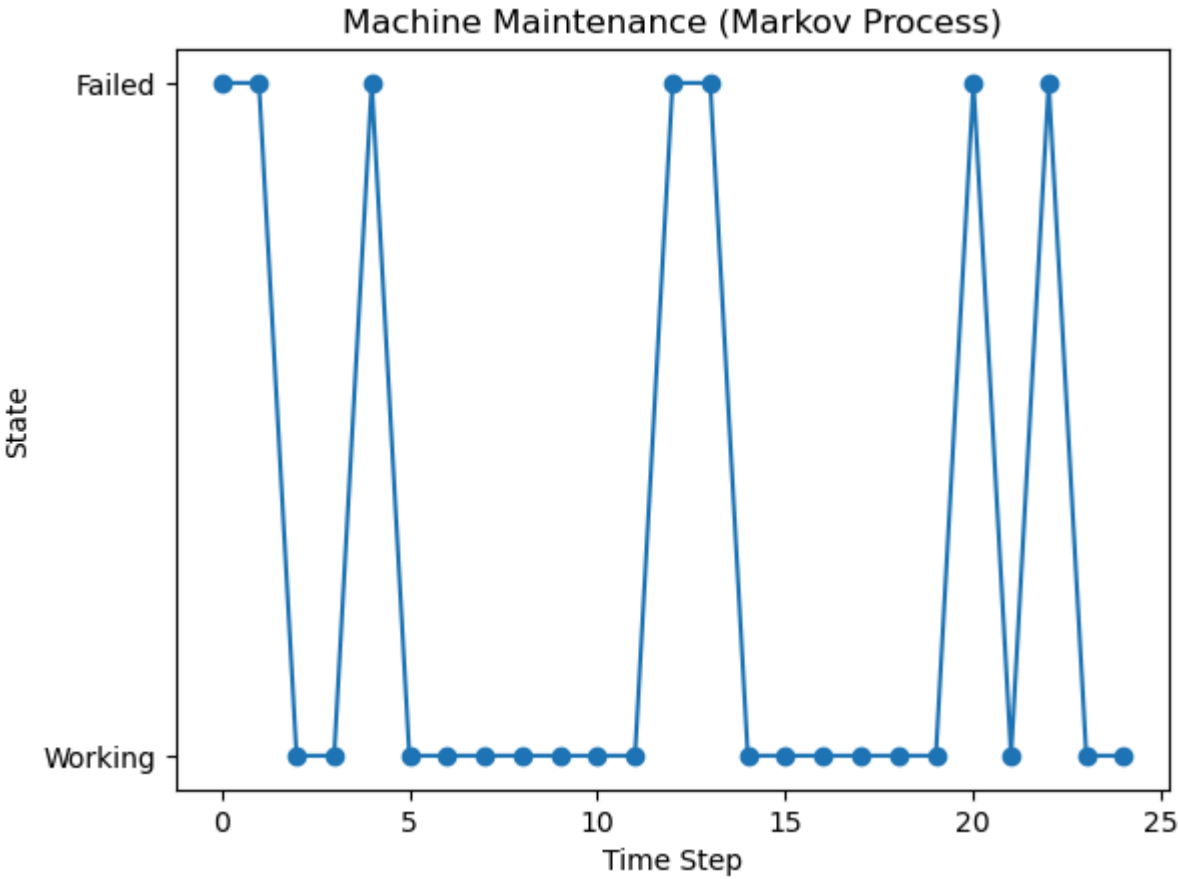


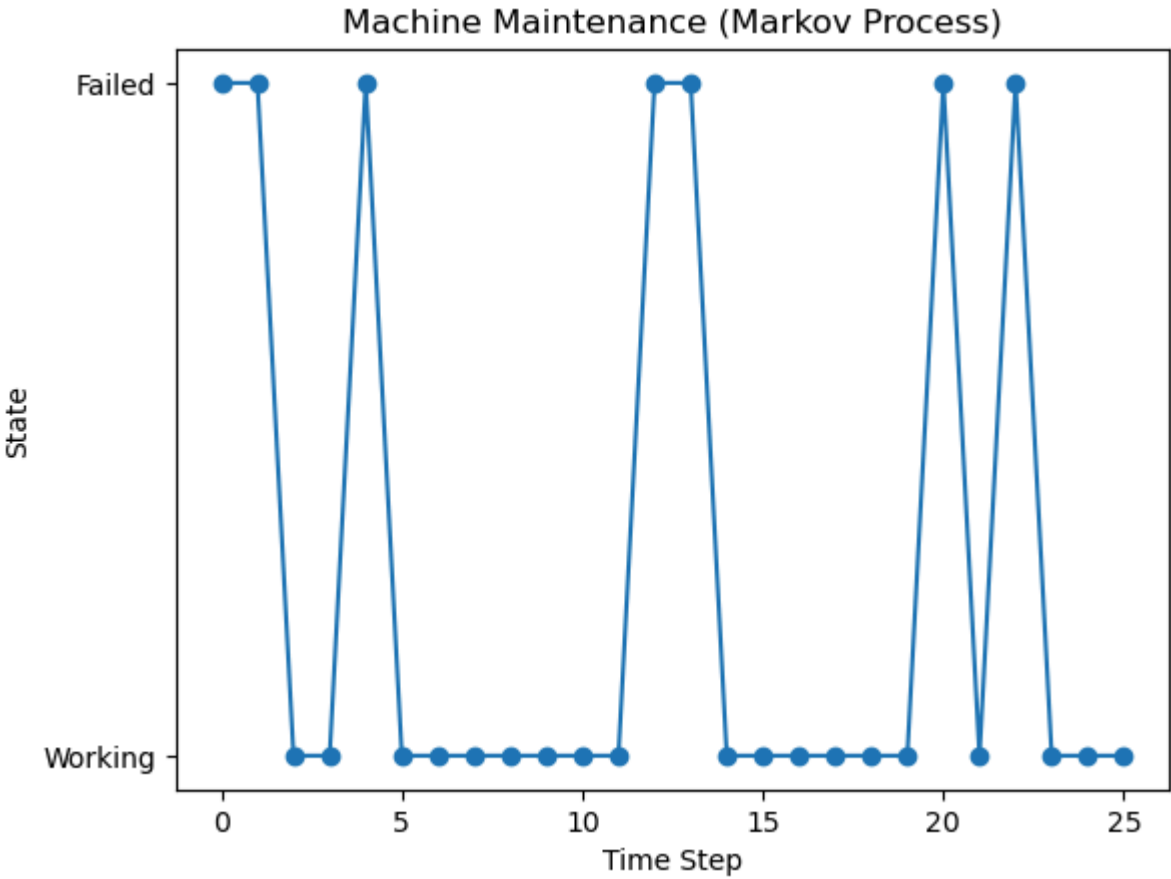


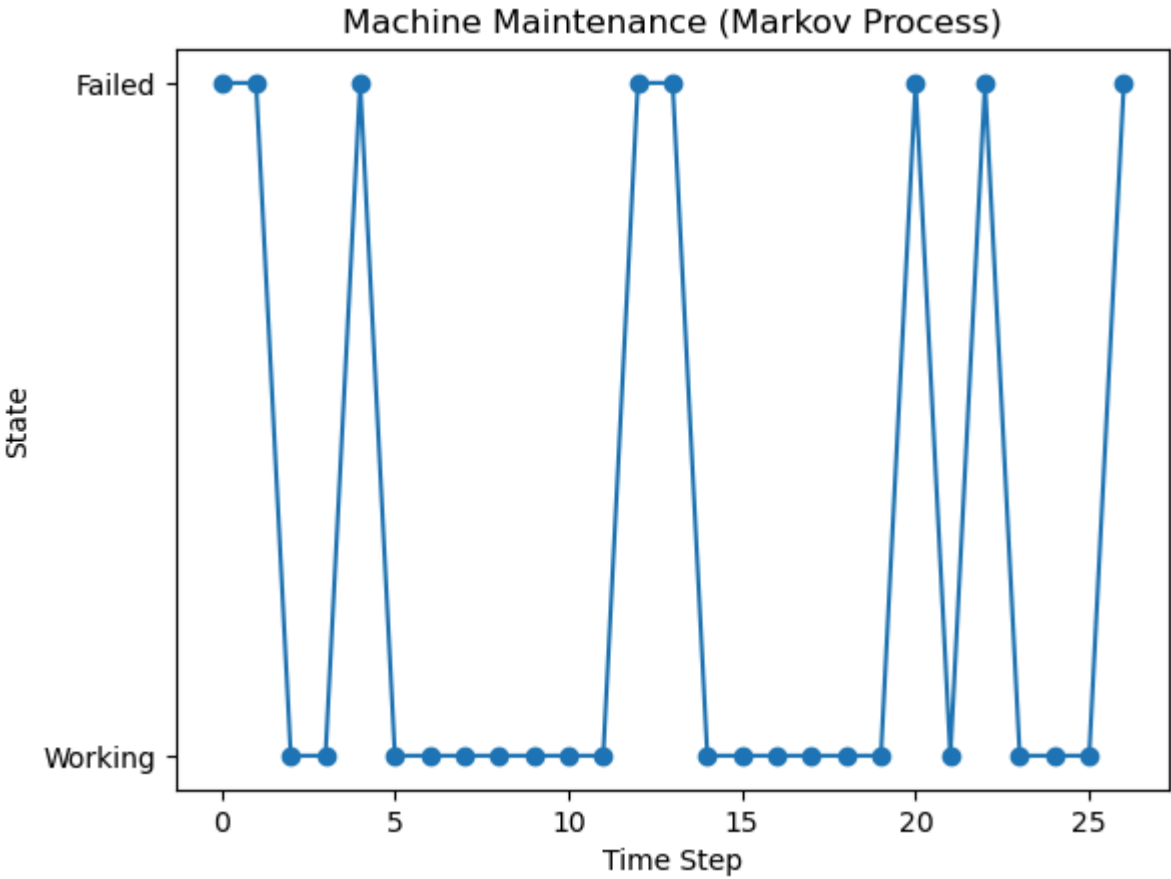


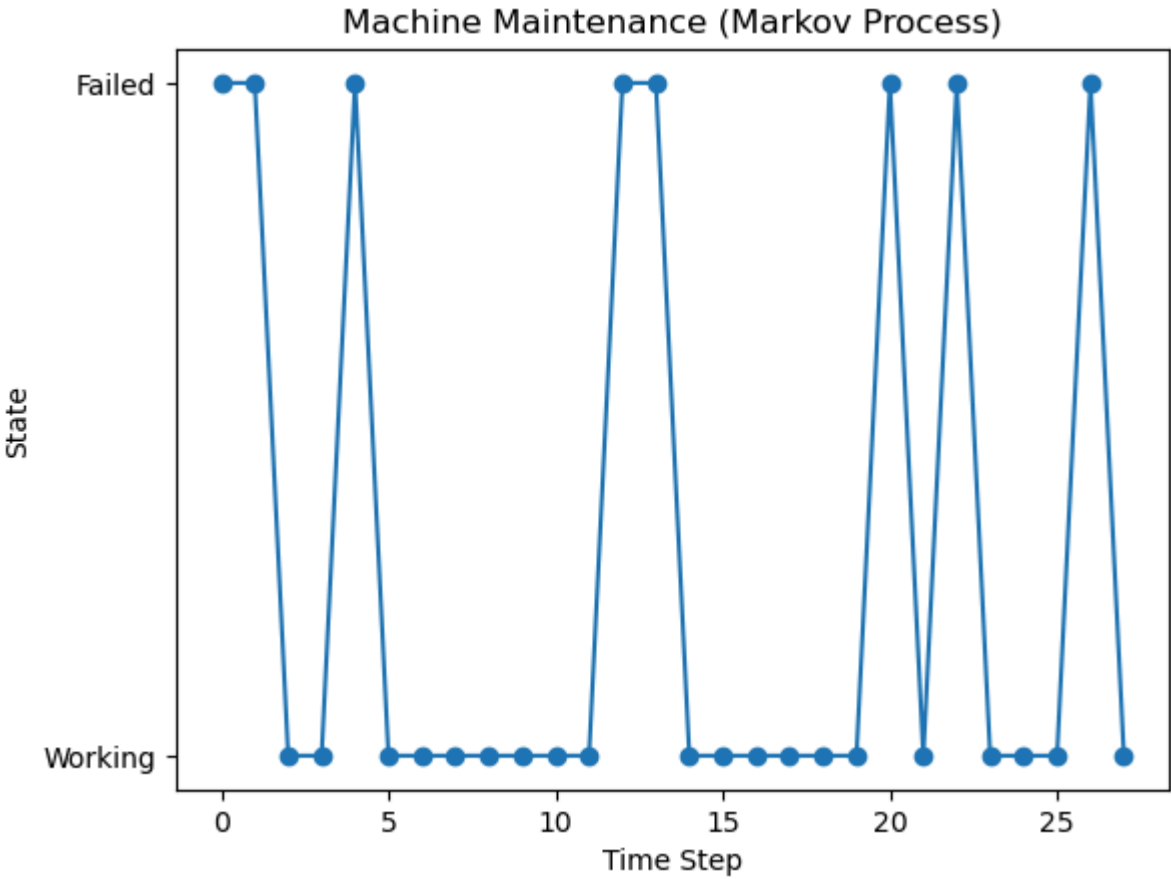


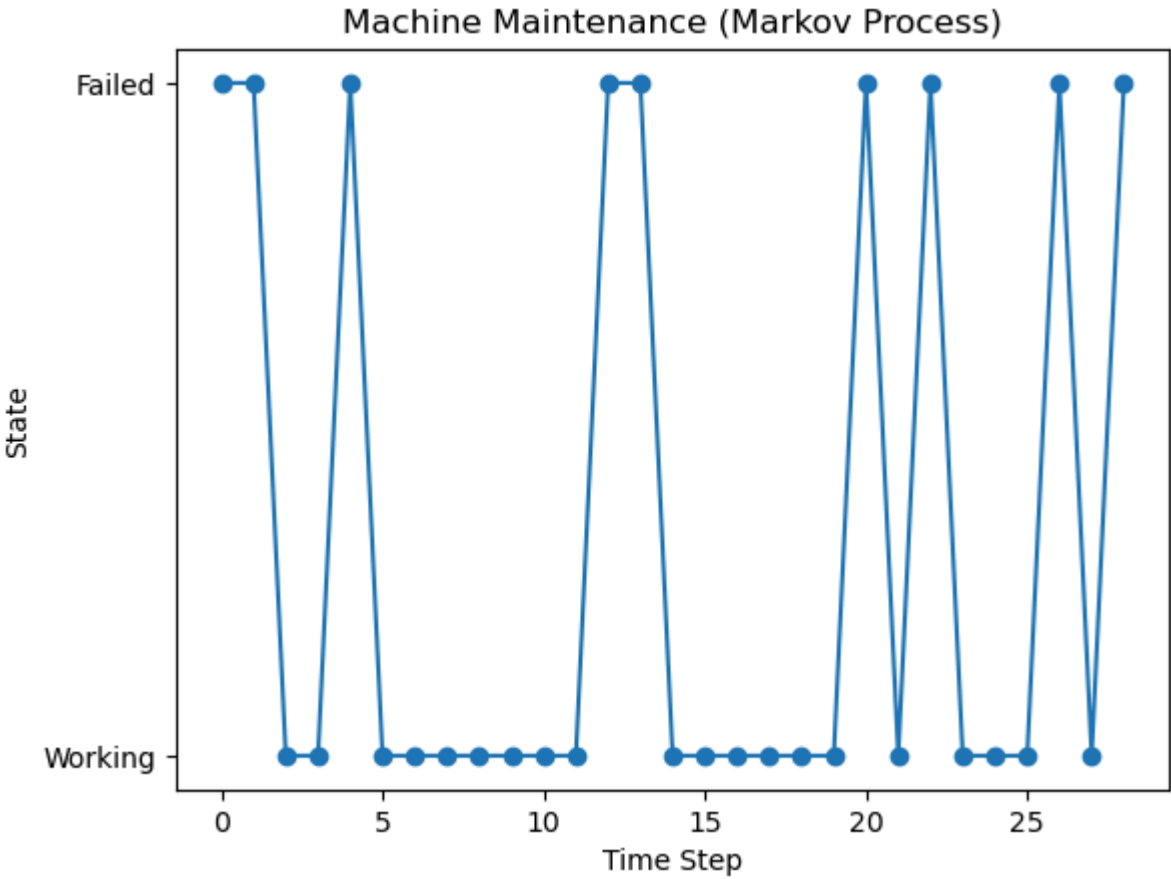


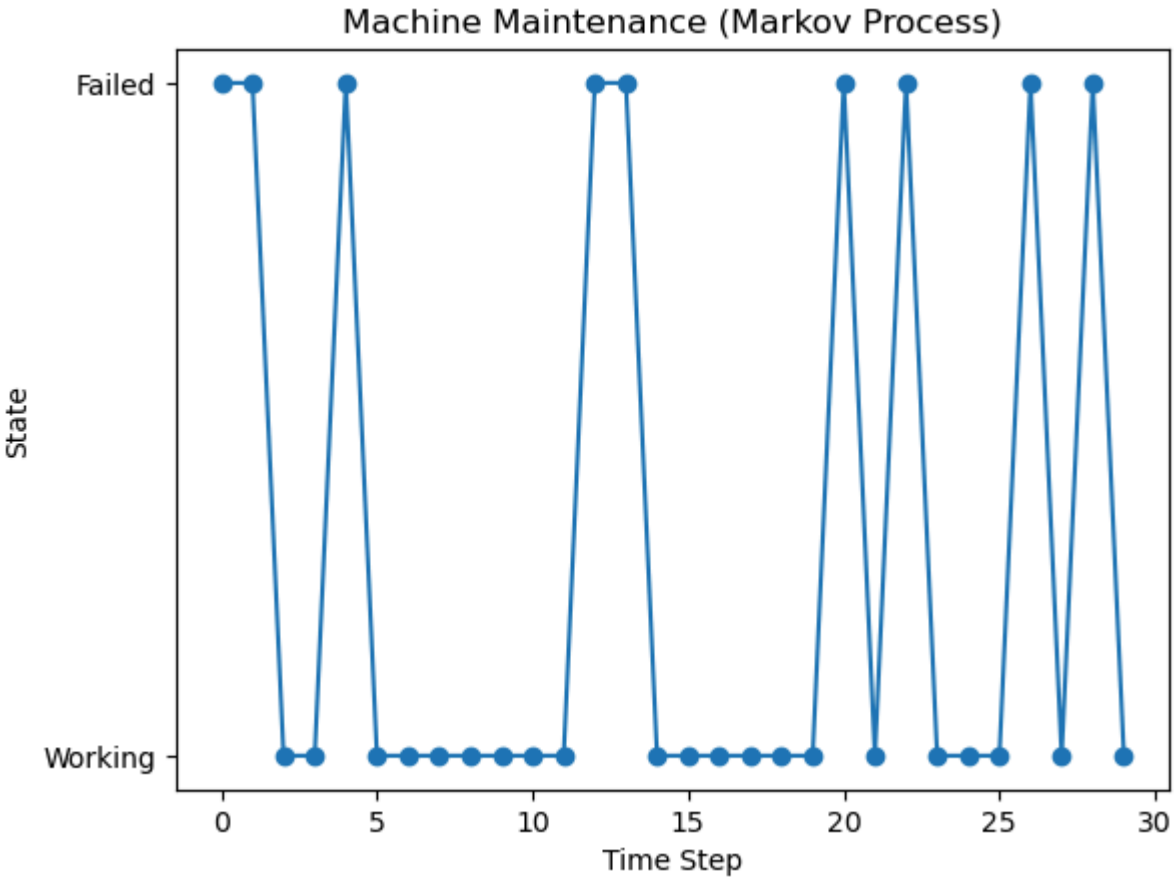


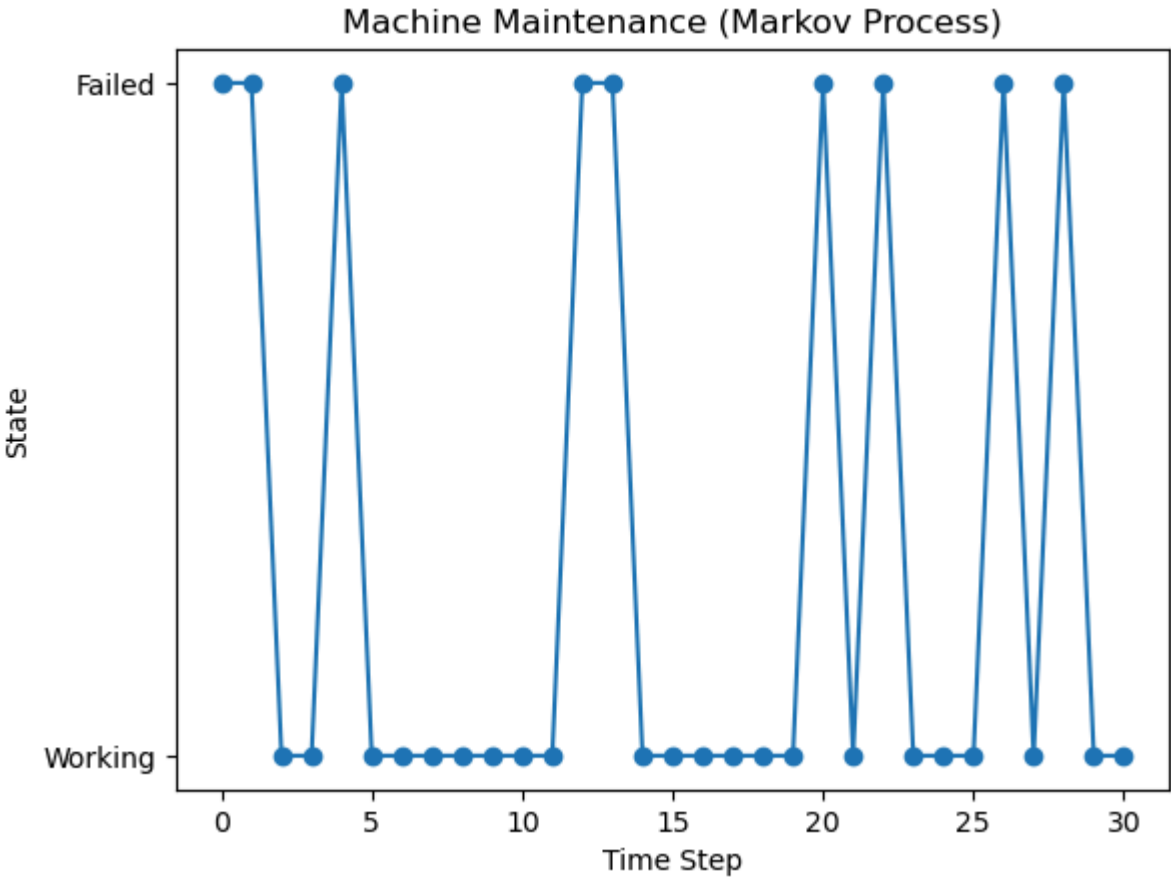


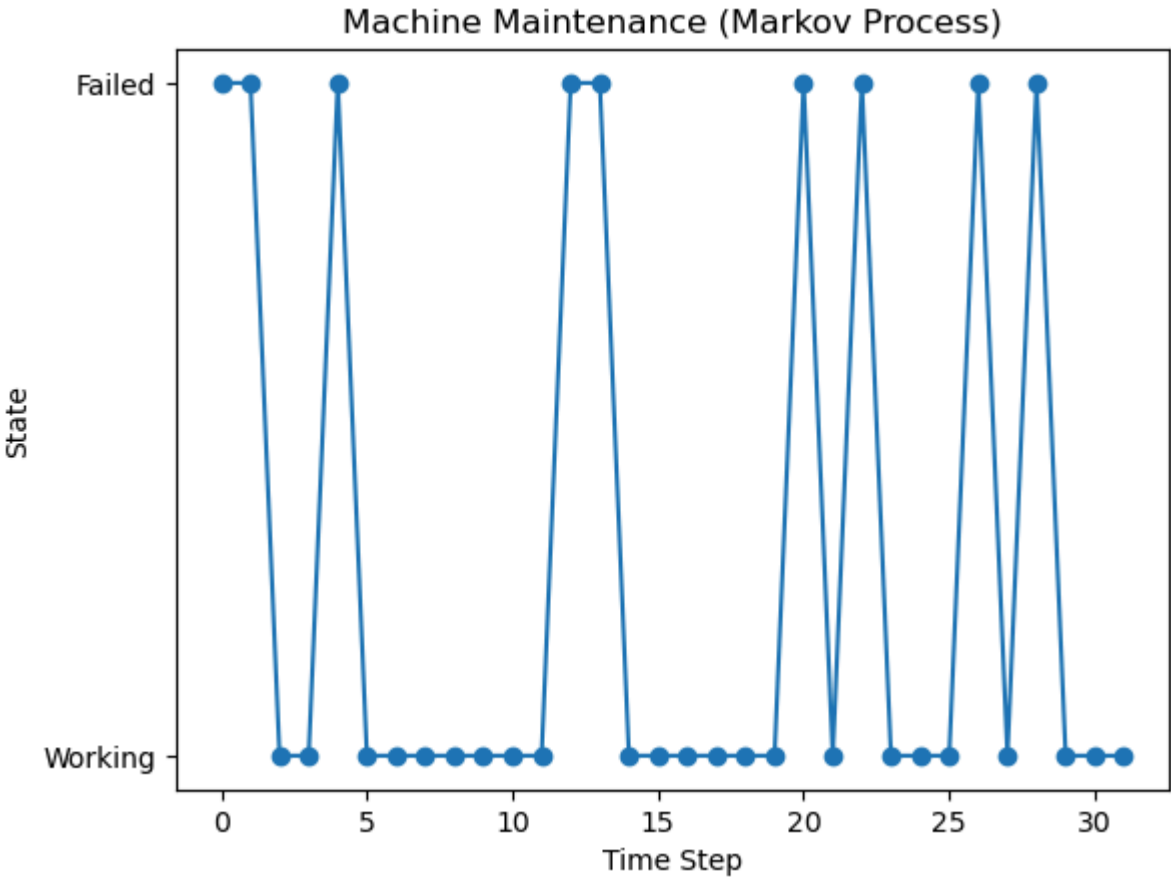


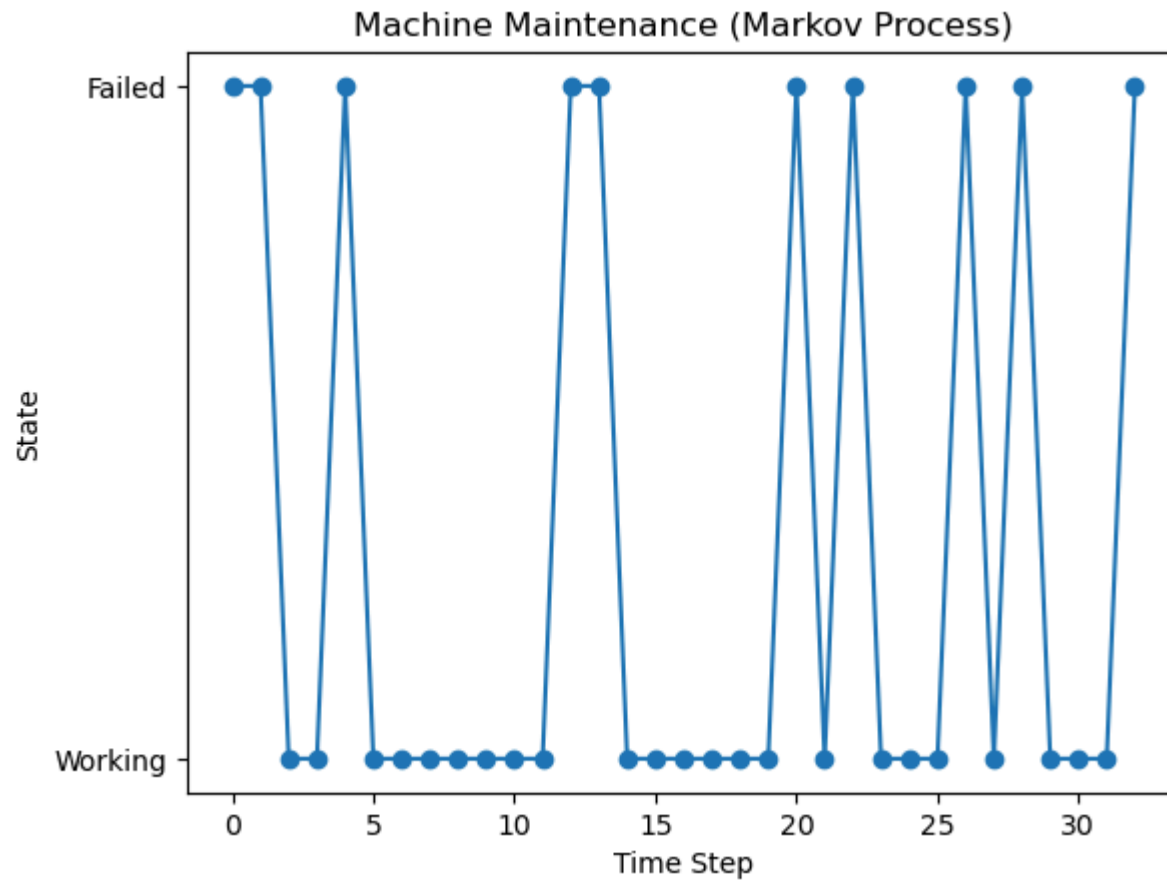


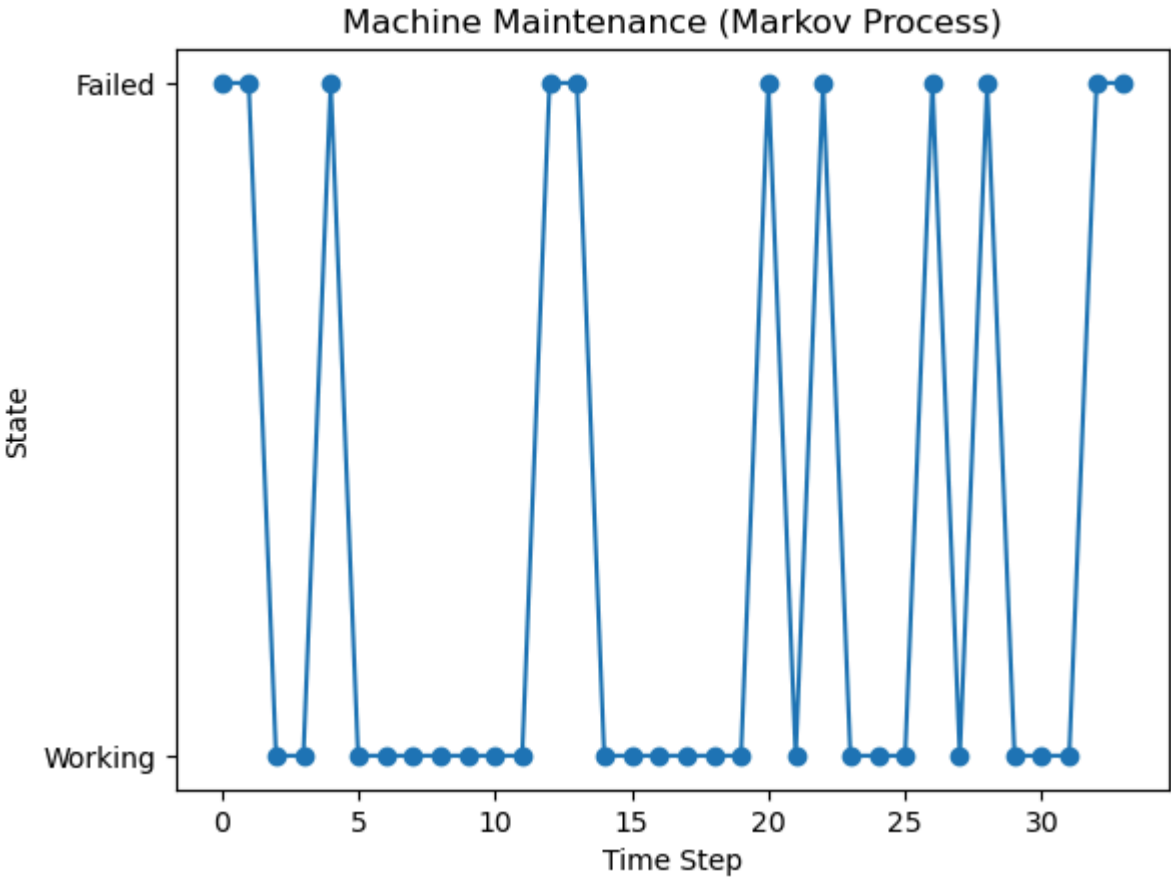


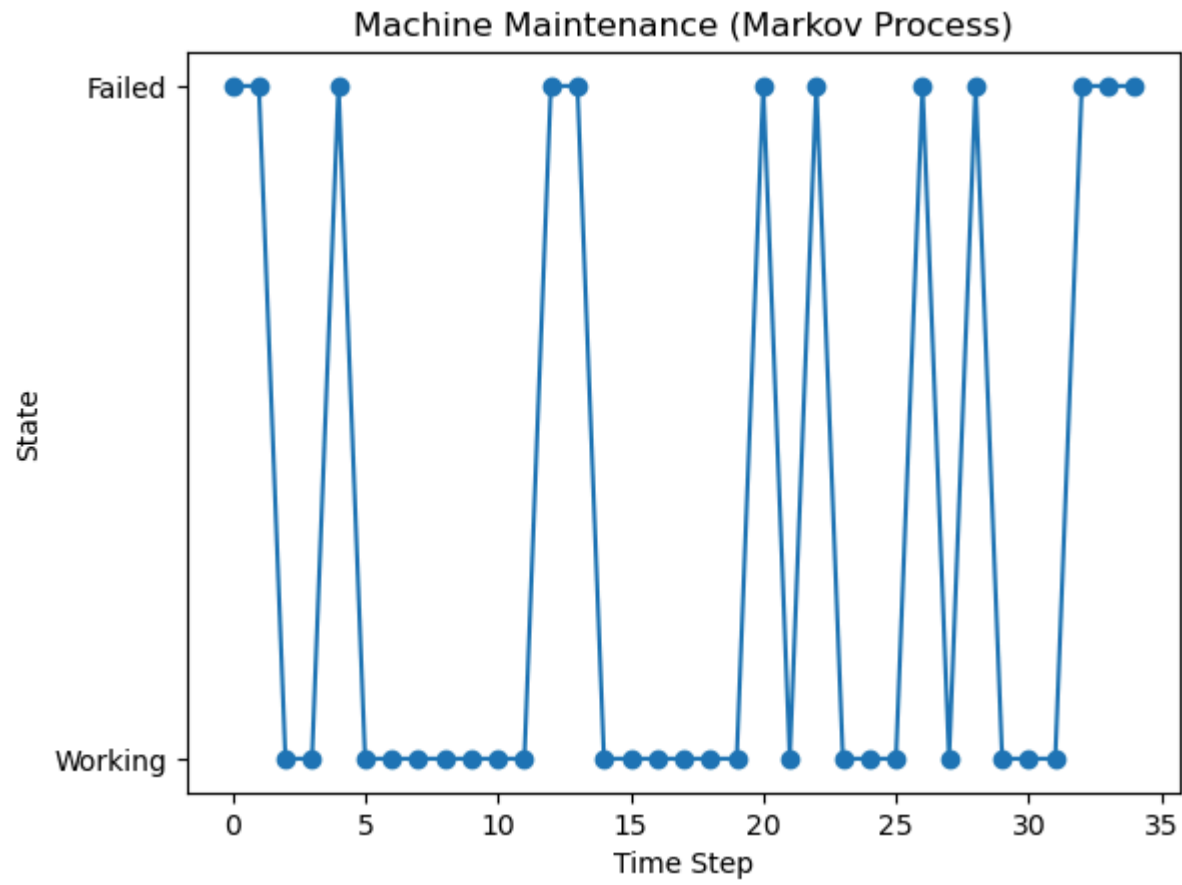


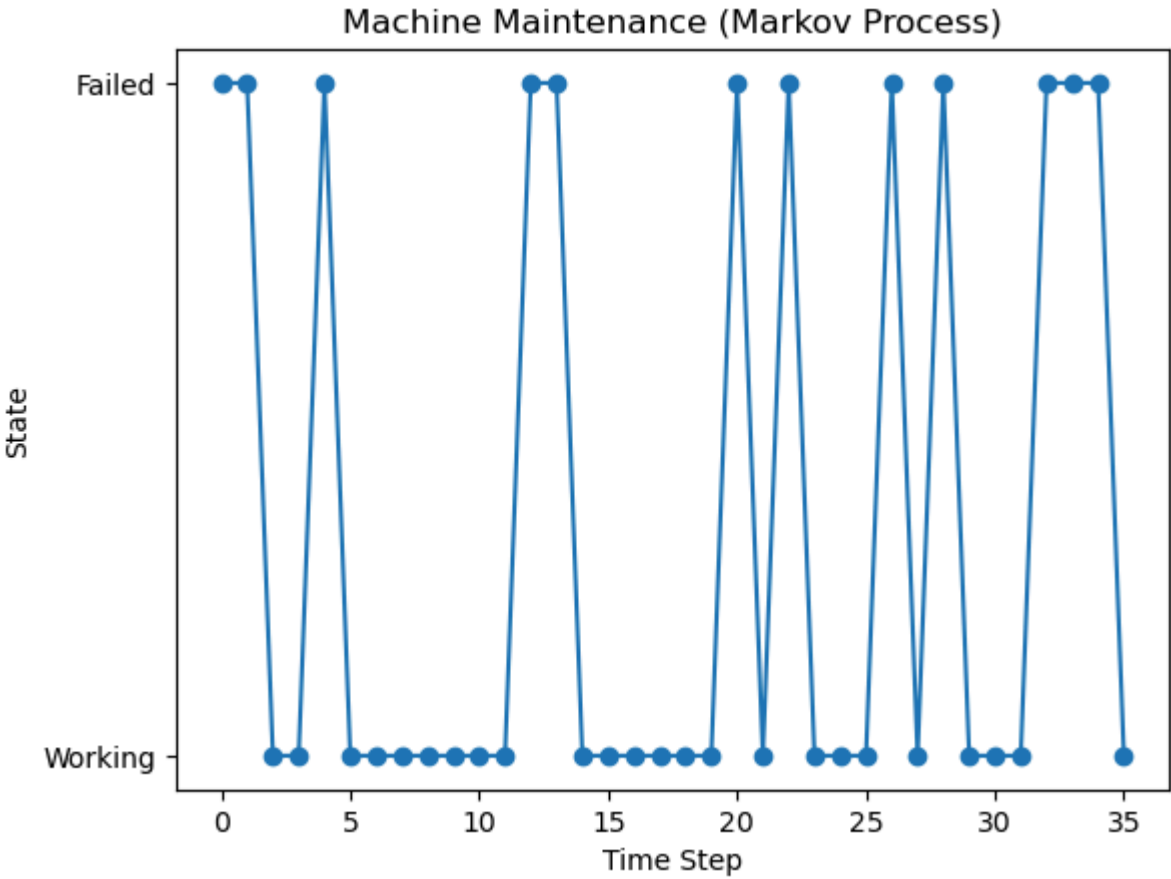


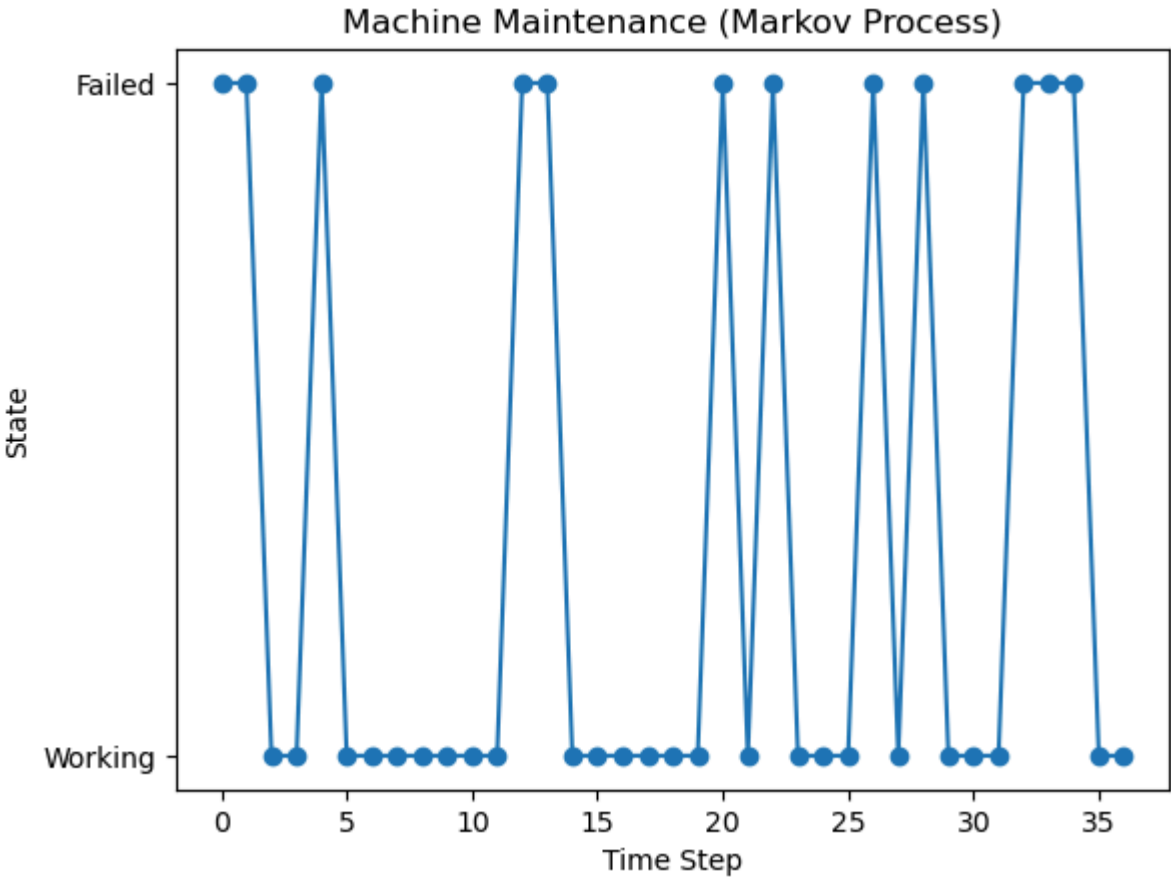


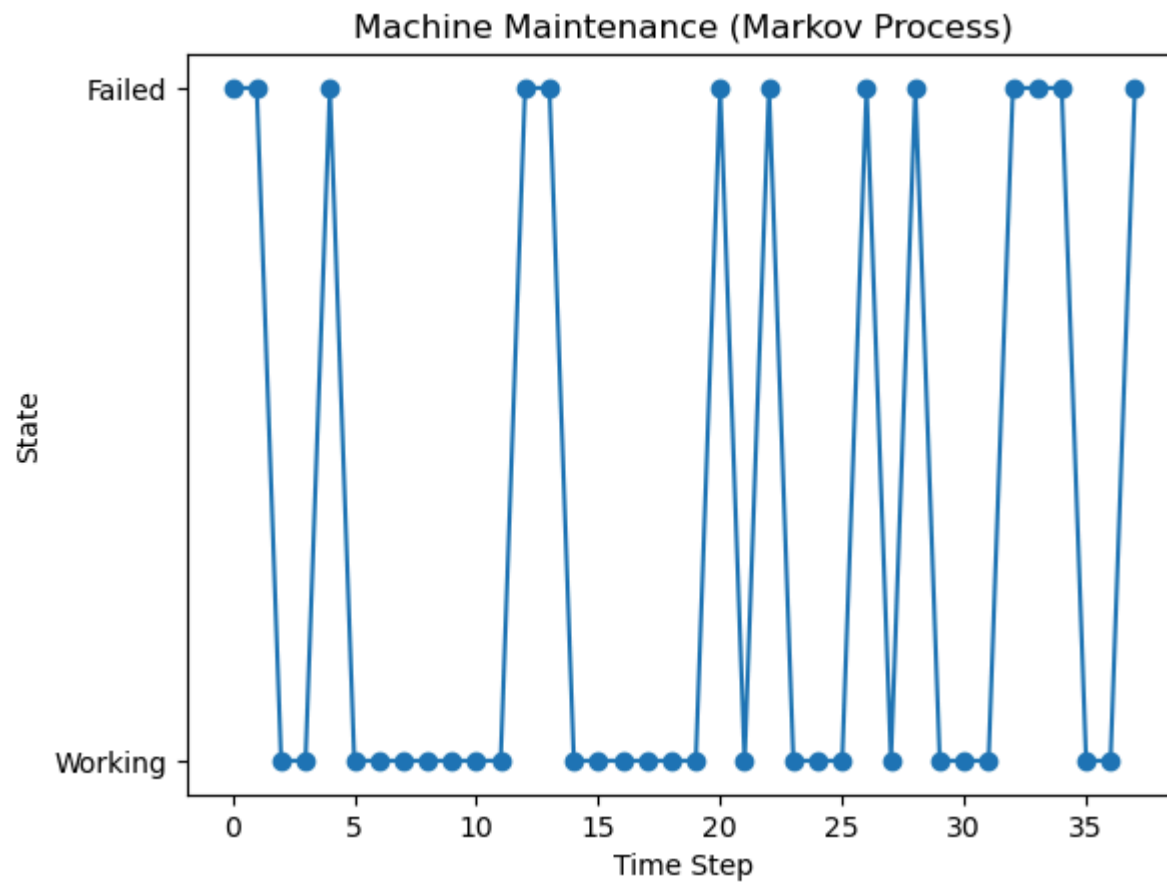


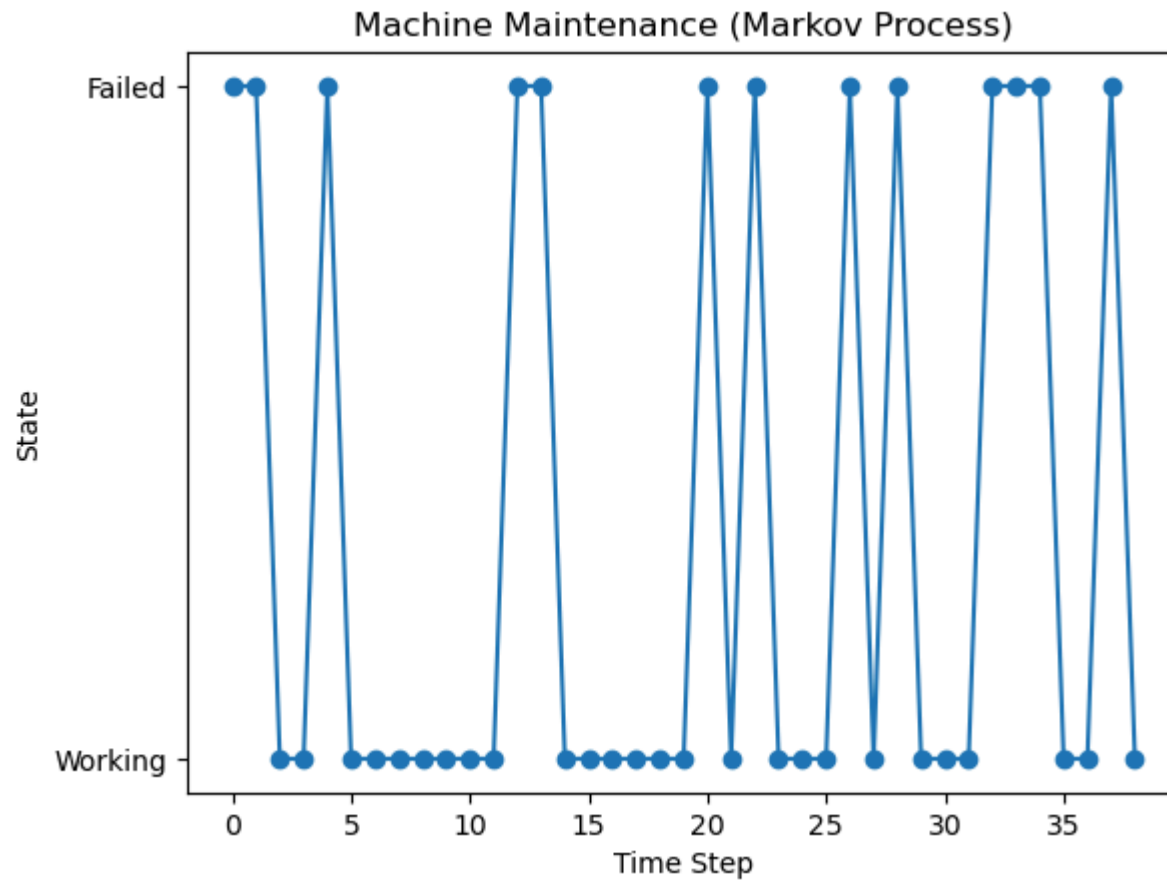


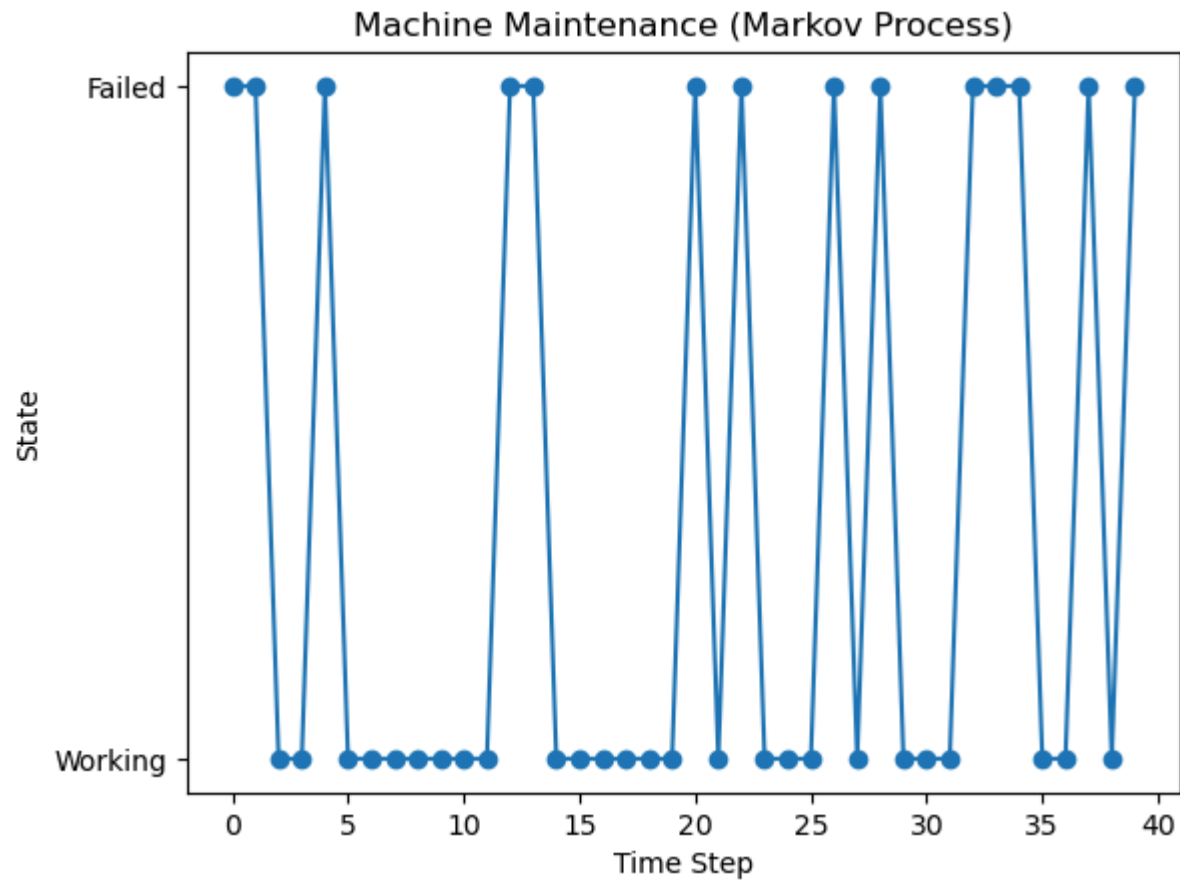


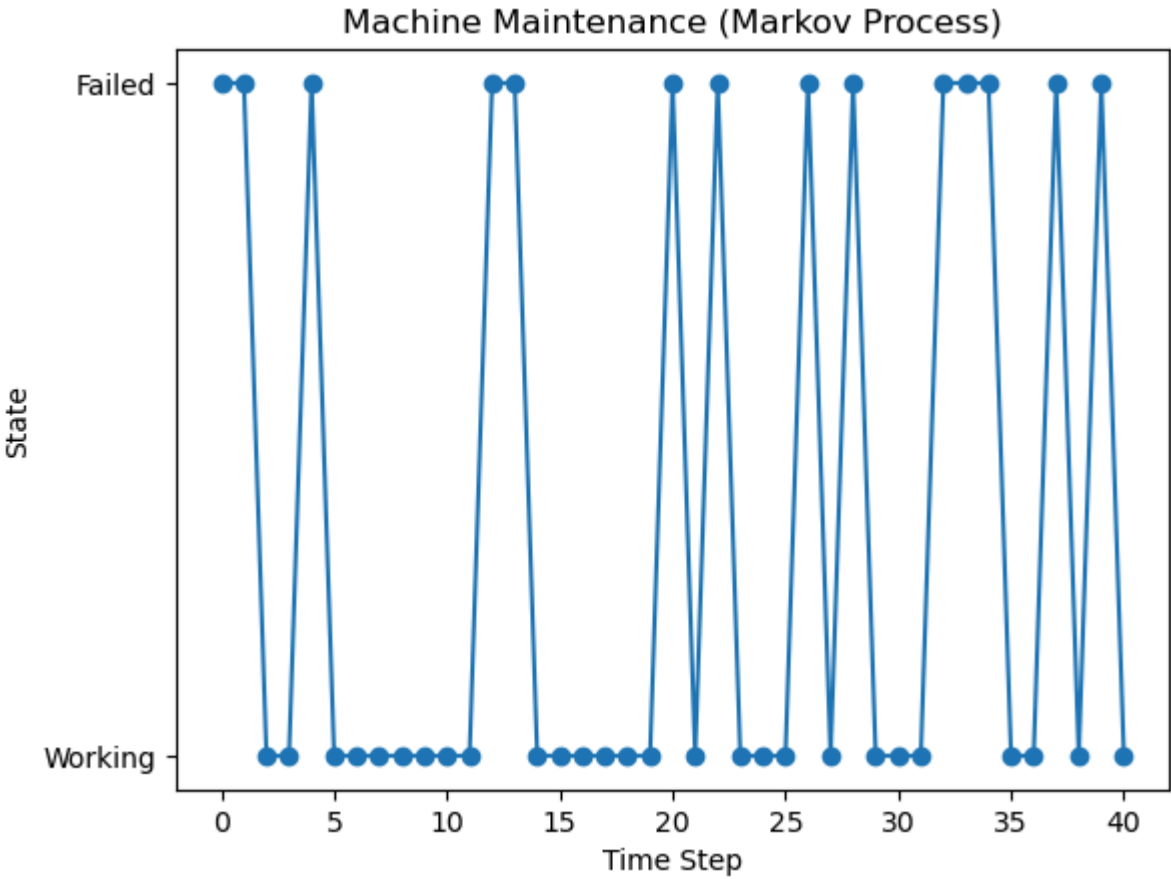


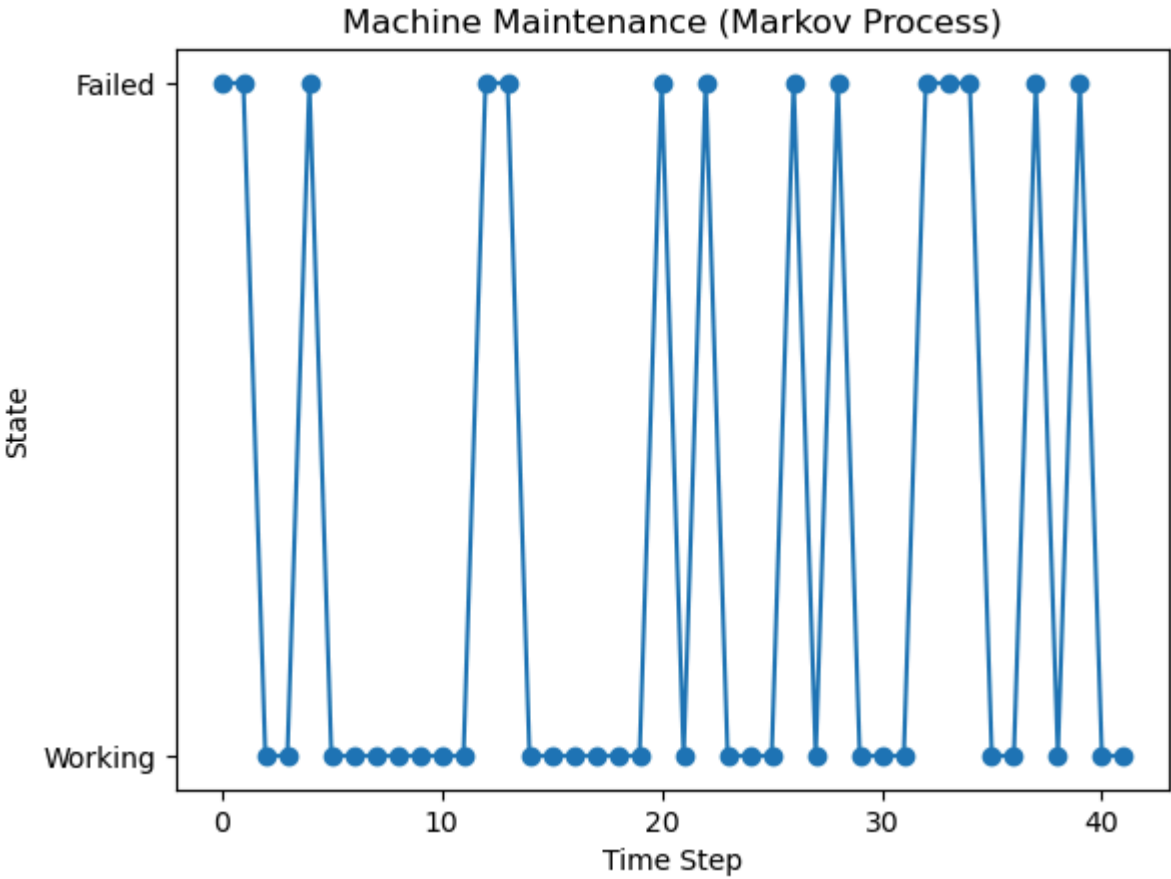


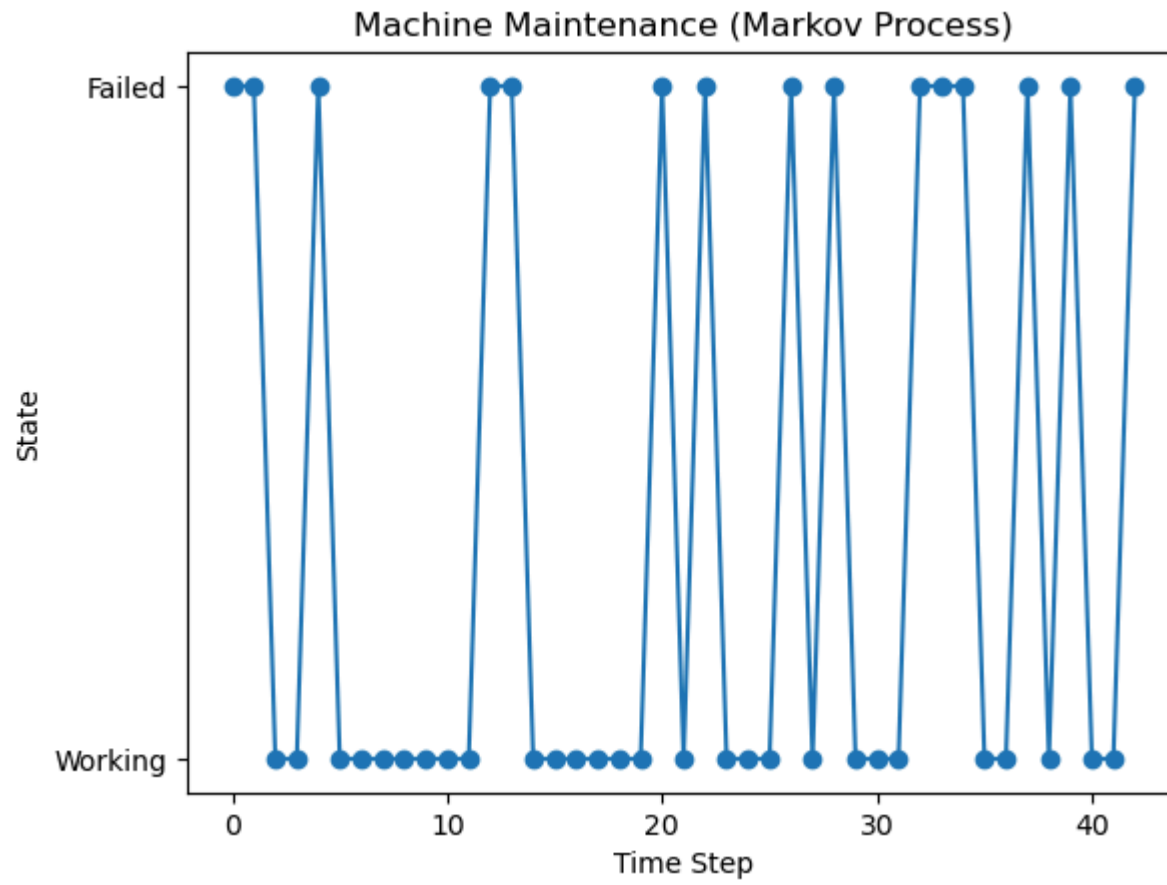


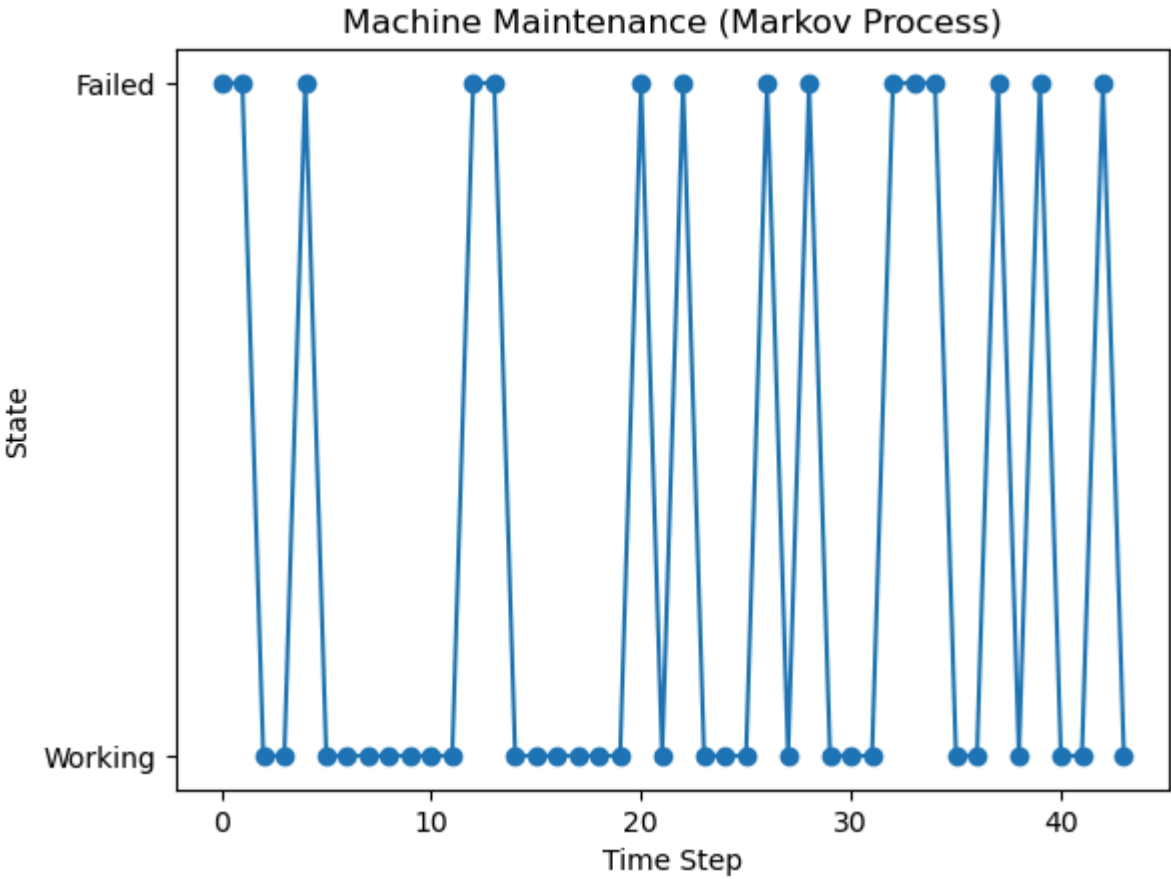


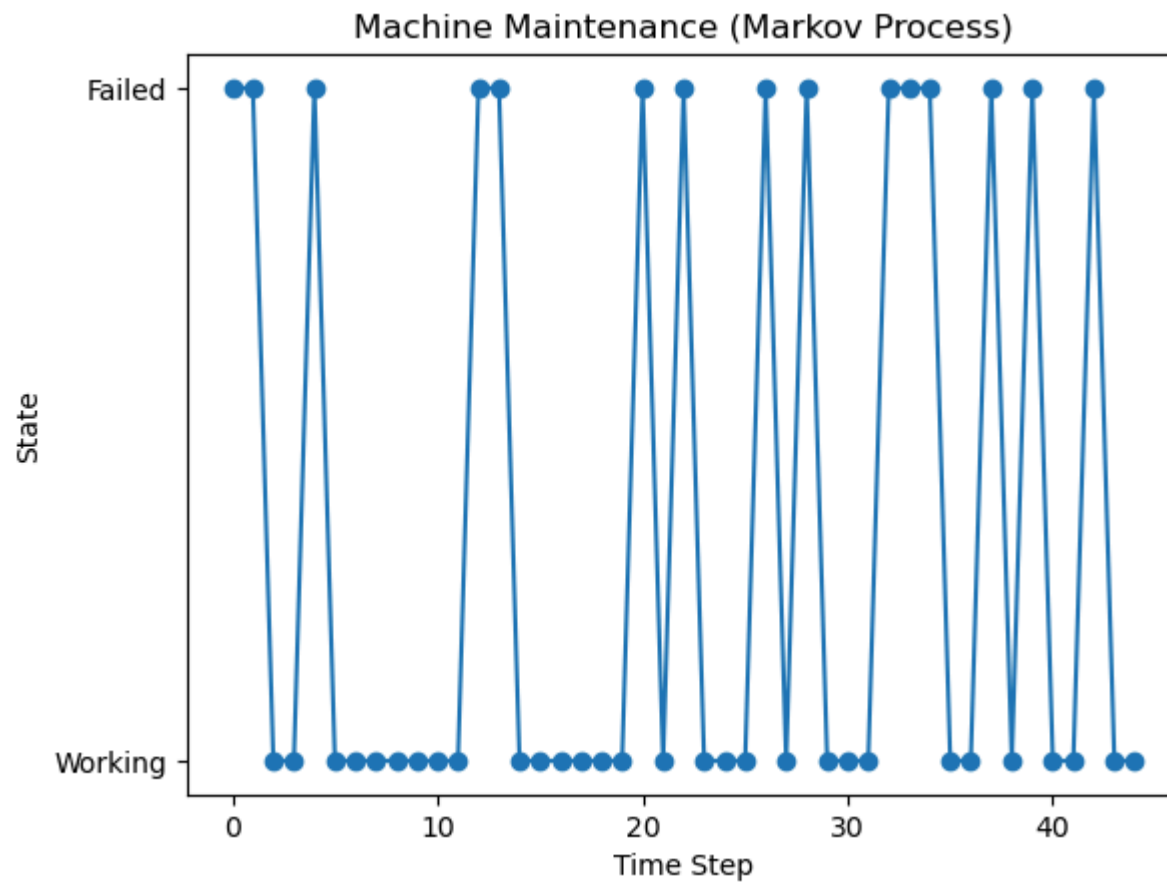


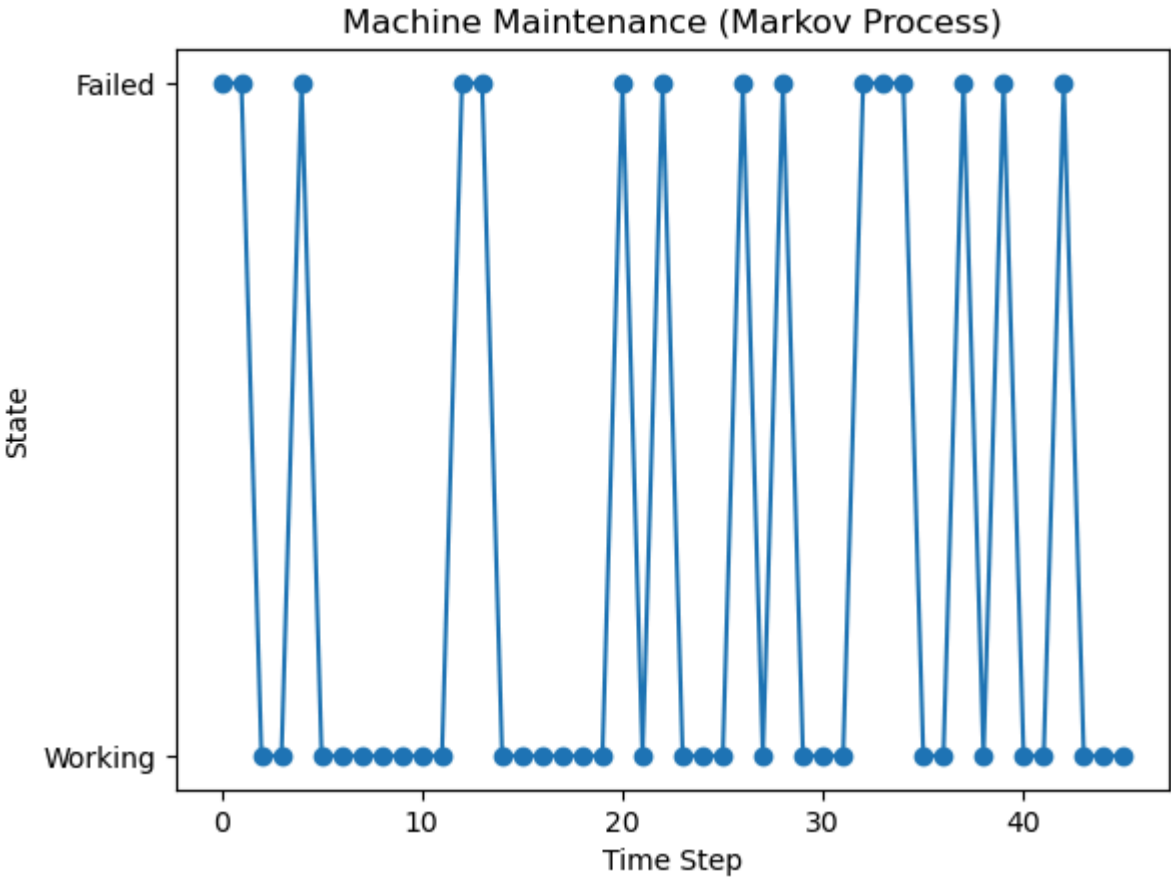


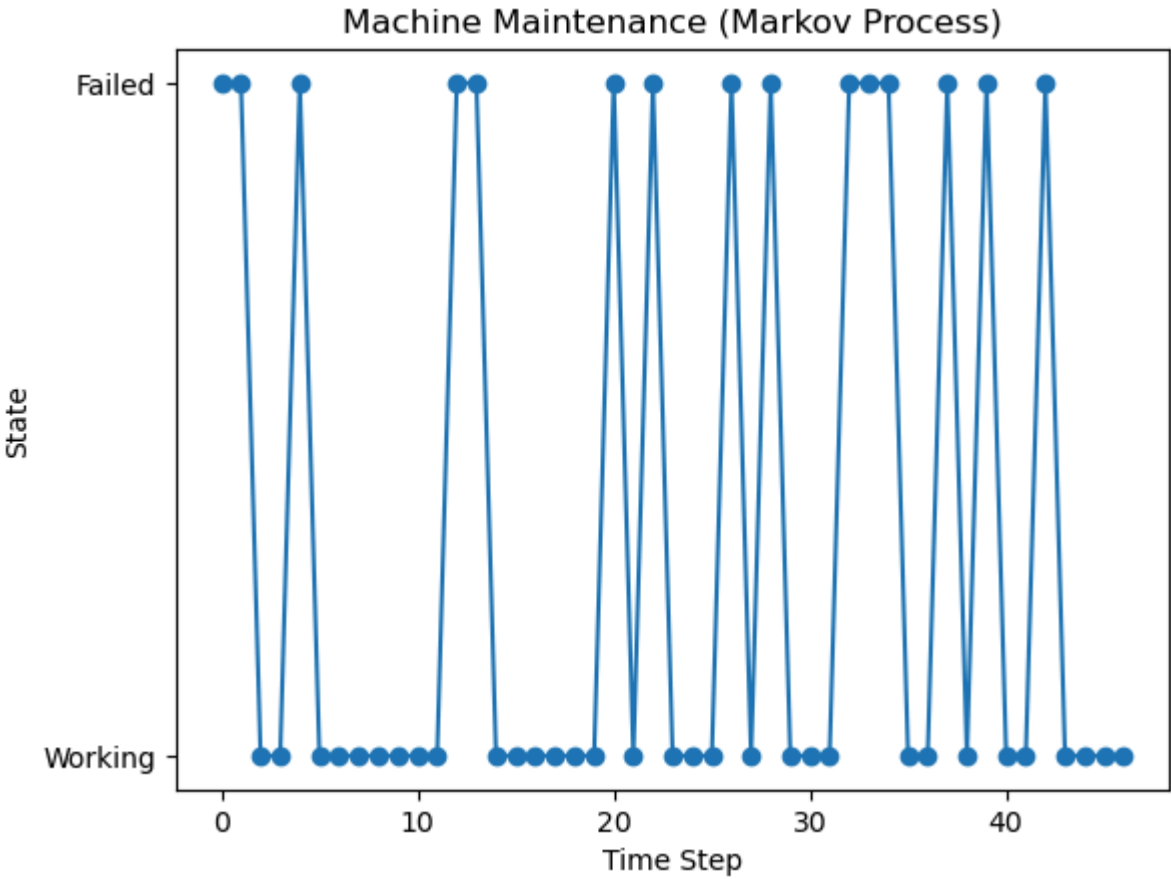


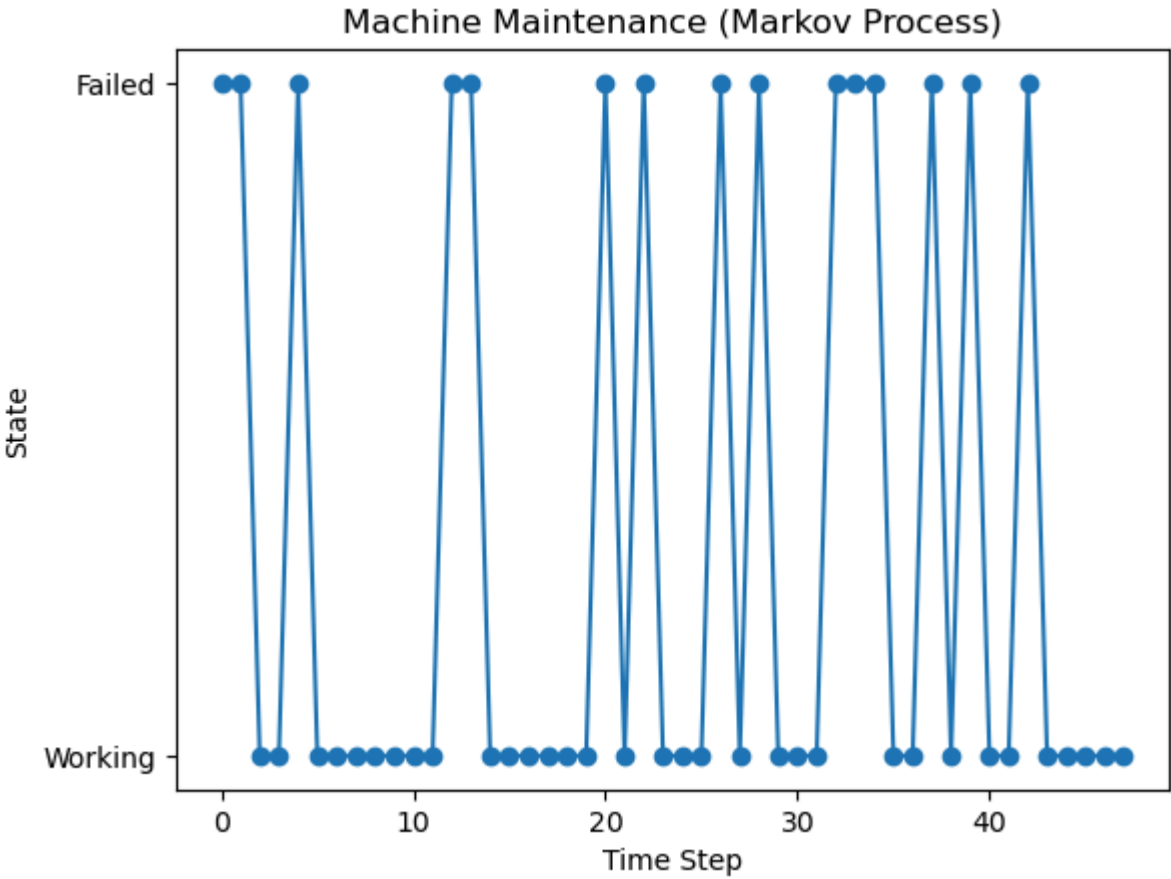


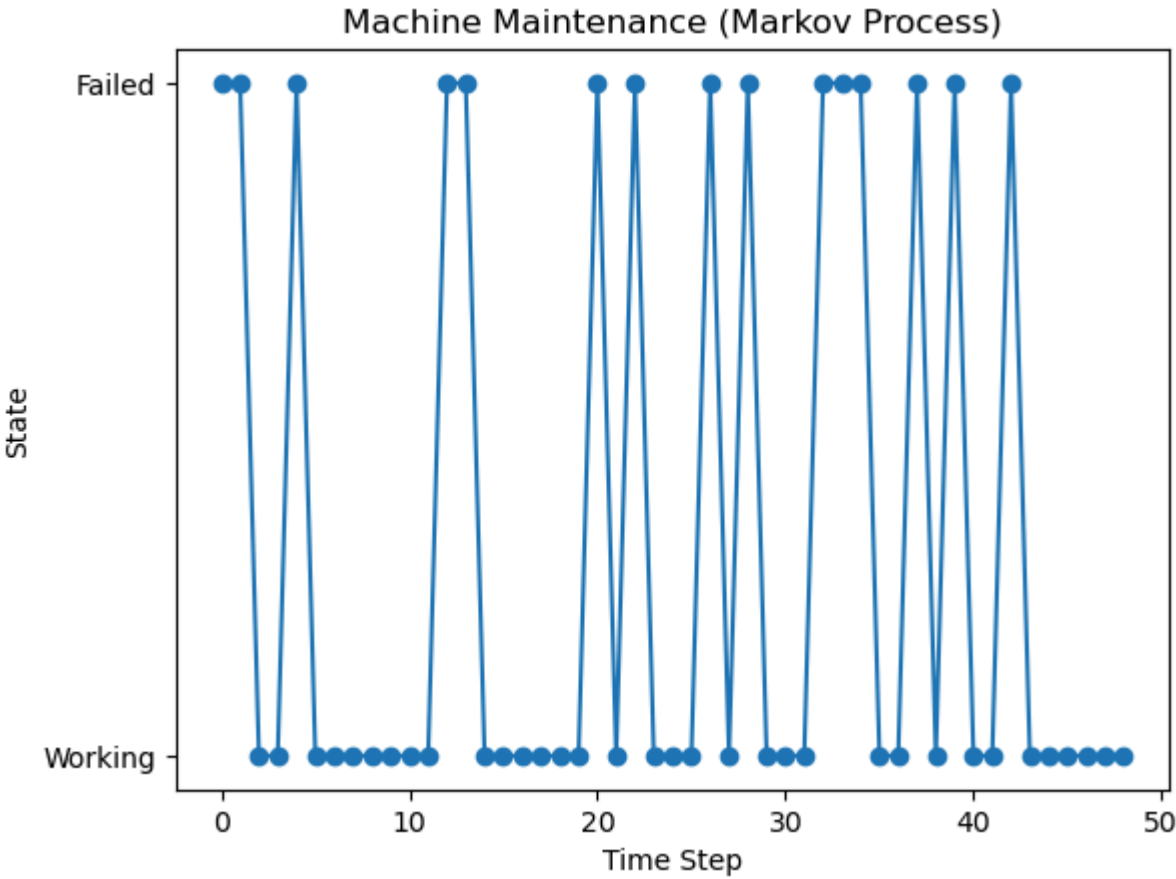


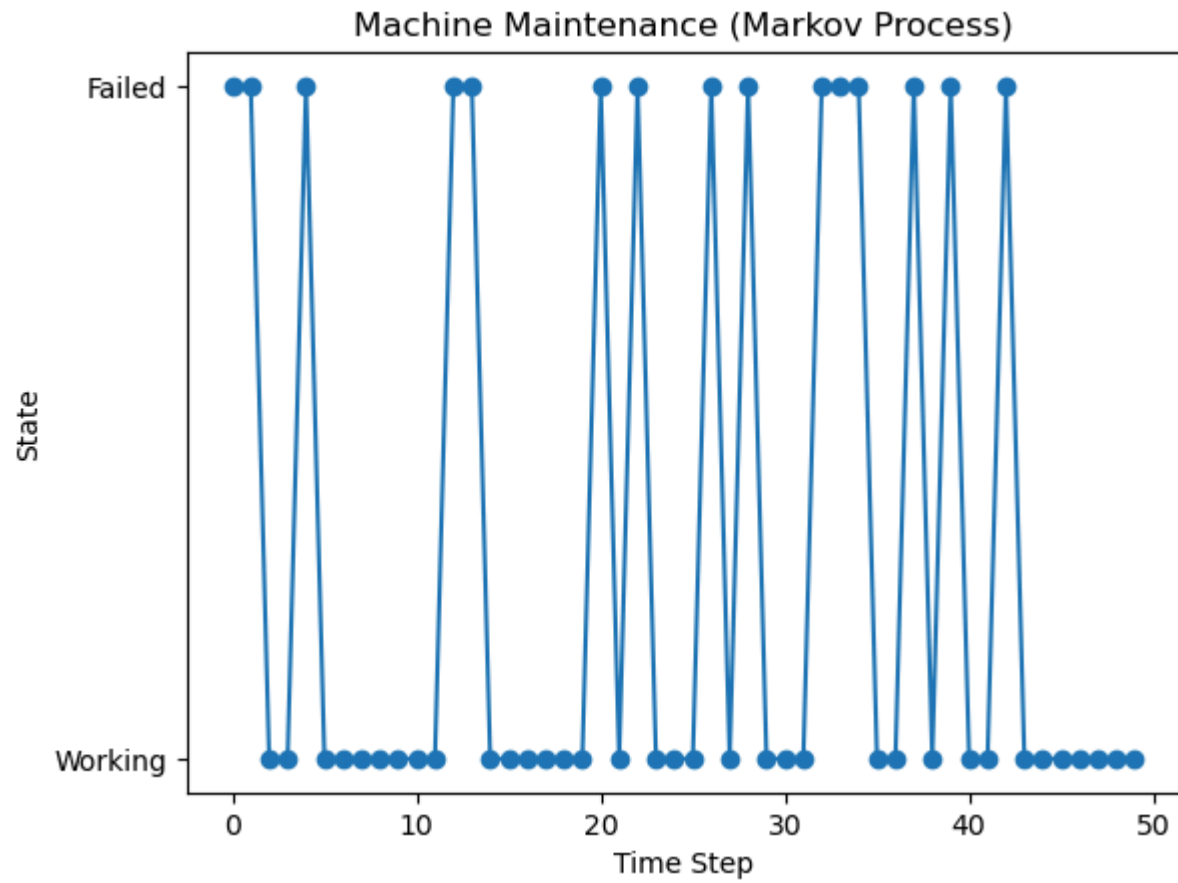












In []: